

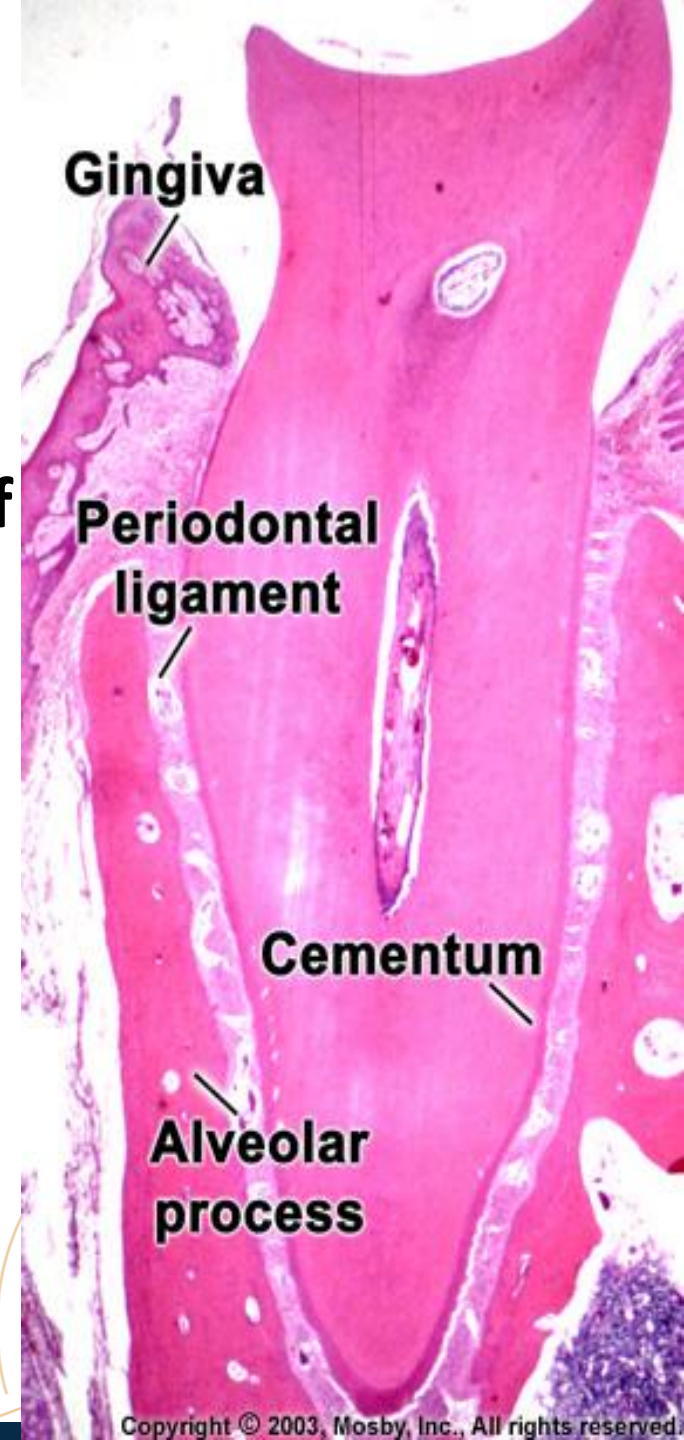
Cementum



a specialized C.T. with calcified inter-cellular substance that covers the anatomical root.

- It develops from the cementoblasts that are differentiated from the UMC of the dental sac during root formation.

It acts as a medium for the attachment of collagen fibers of periodontal ligament (**Sharpey's fibers**) that bind the tooth to the alveolar bone.



Physical Characteristics

1-Color

★ Light yellow

2- Thickness

Acellular cementum (20-50 μ)

Cellular cementum (150-200 μ)

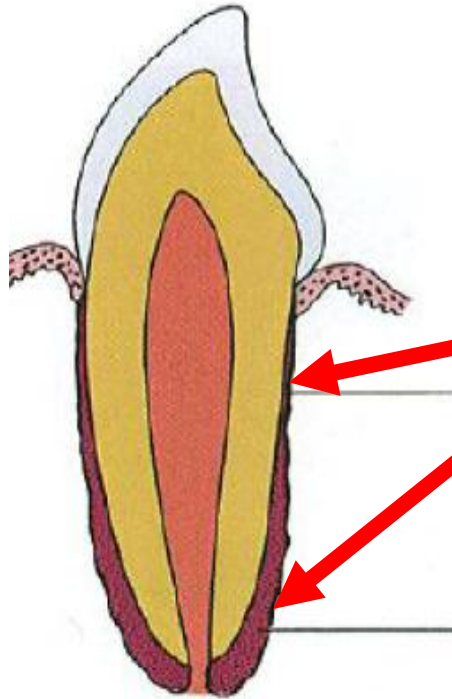
3- Permeability

Permeable from dentin and PDL sides.

Cellular C is more permeable than acellular C.

4-Hardness

Less than dentine



Chemical Composition

45-50 % Inorganic
substances



Hydroxyapatite
crystals

50-55% Organic
substances

Collagen



protein

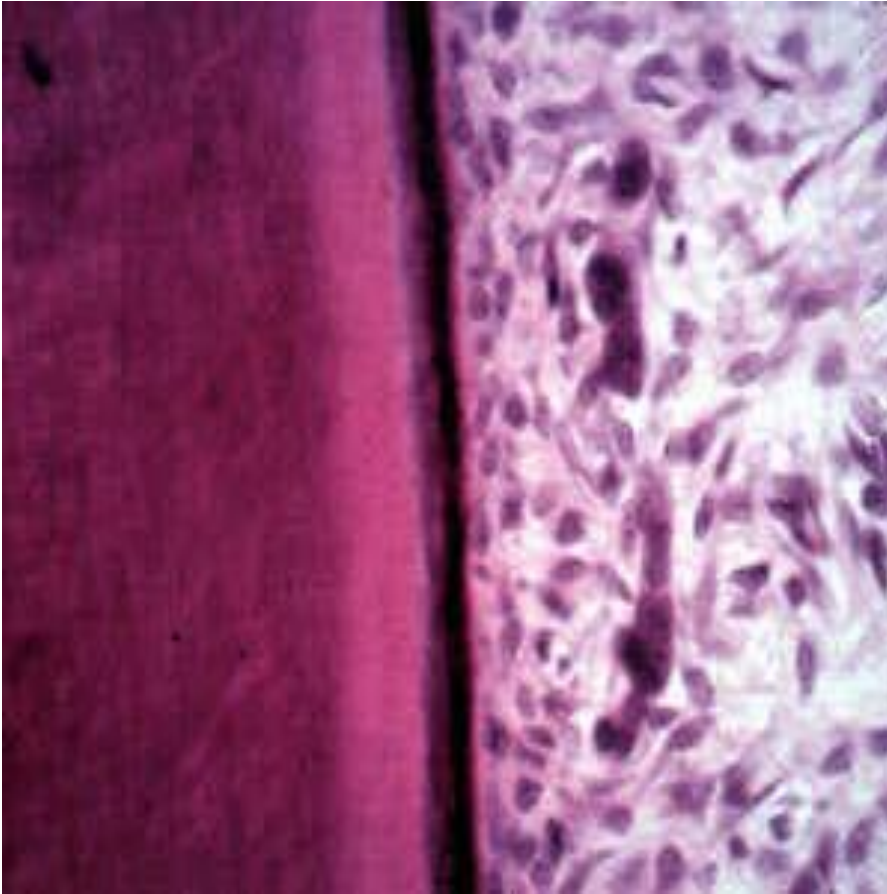


Polysaccharides

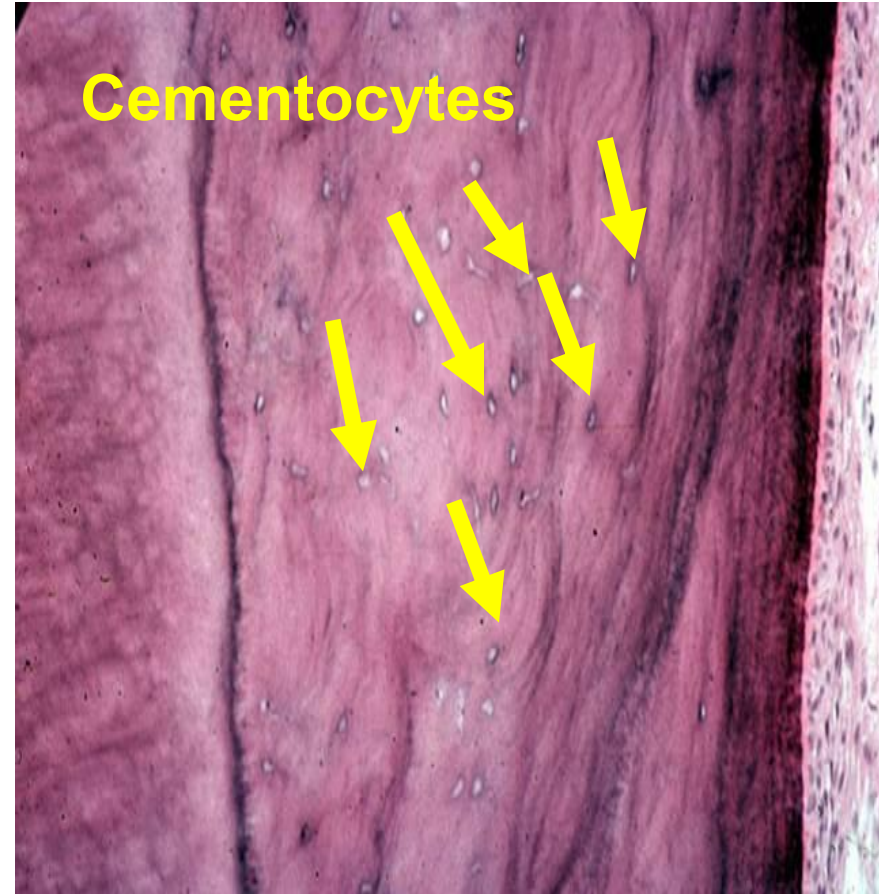


**Cementum contains the greatest amount of
fluoride in all mineralized tissues**

Cementum Structure

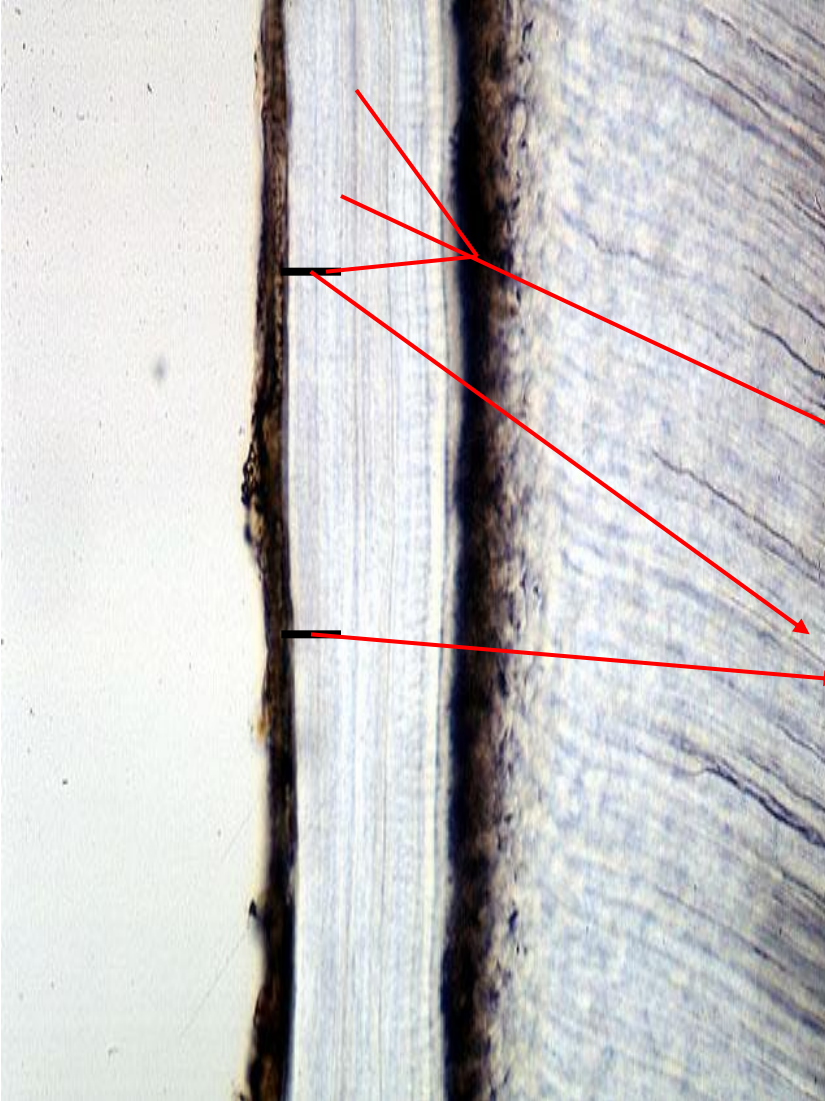


Acellular cementum



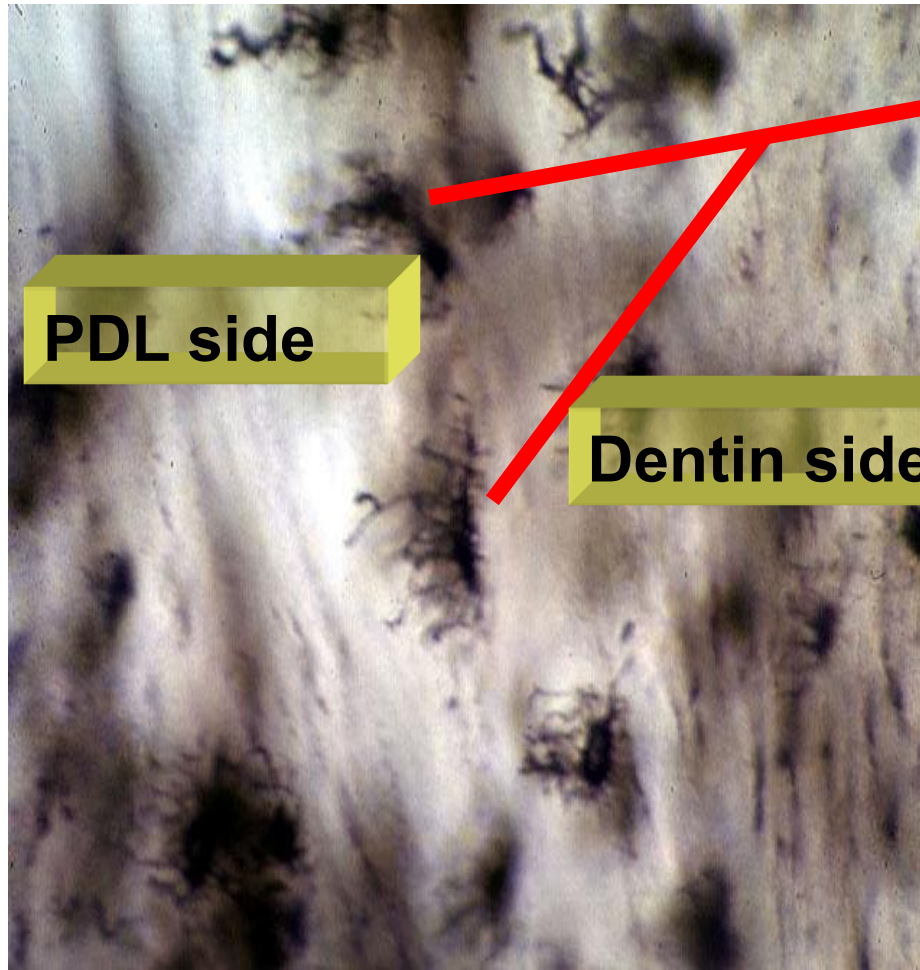
Cellular cementum

1-Acellular Cementum

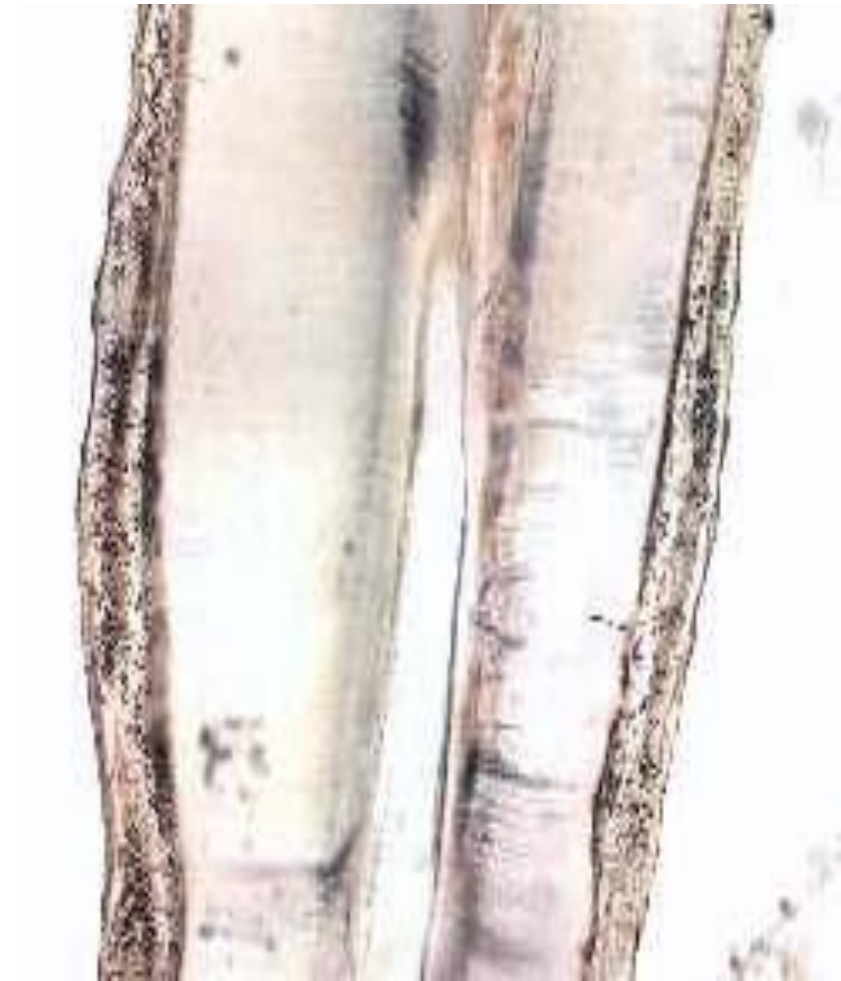


- Thickness is 20-50 μ .
 - Free of cells ,contains collagen fibers & ground substance.
 - It is clear and structure less
 - Covers the cervical and middle thirds of the root.
 - Incremental lines of **Salter** are parallel to the surface.
 - Sharpey's fibers of periodontal ligament are inserted in it . In ground section Sharpey's fibers appear as dark lines due to its burning.
- Deposition of cementum through out life
- causes calcification of the deeper parts of the Sharpey's fibers.

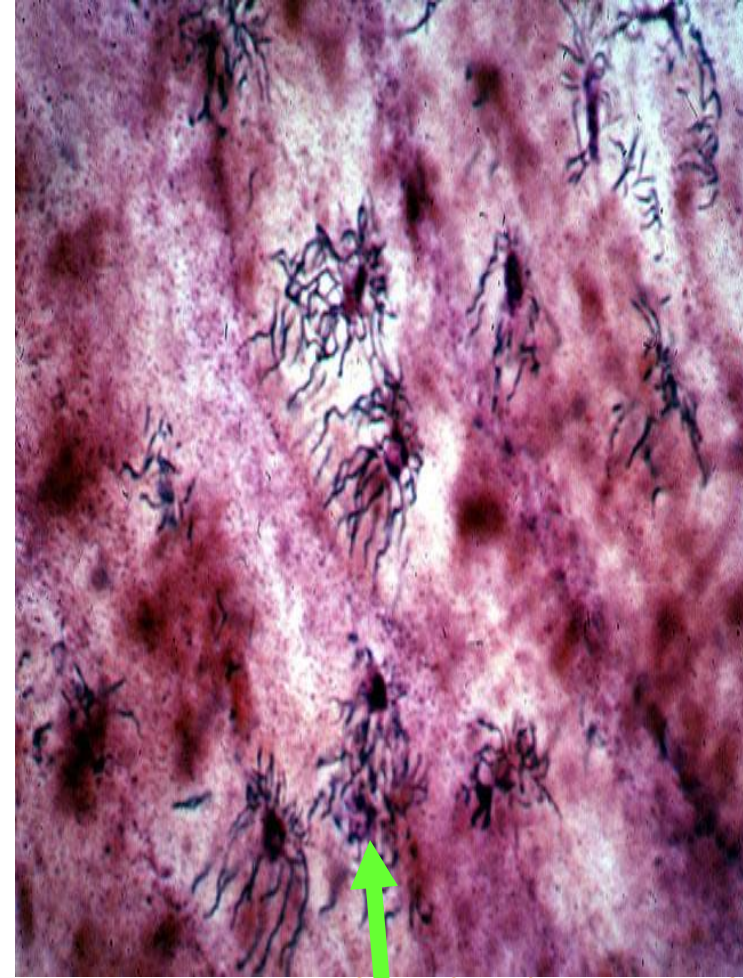
2-Cellular Cementum



Lacunae of cementocytes



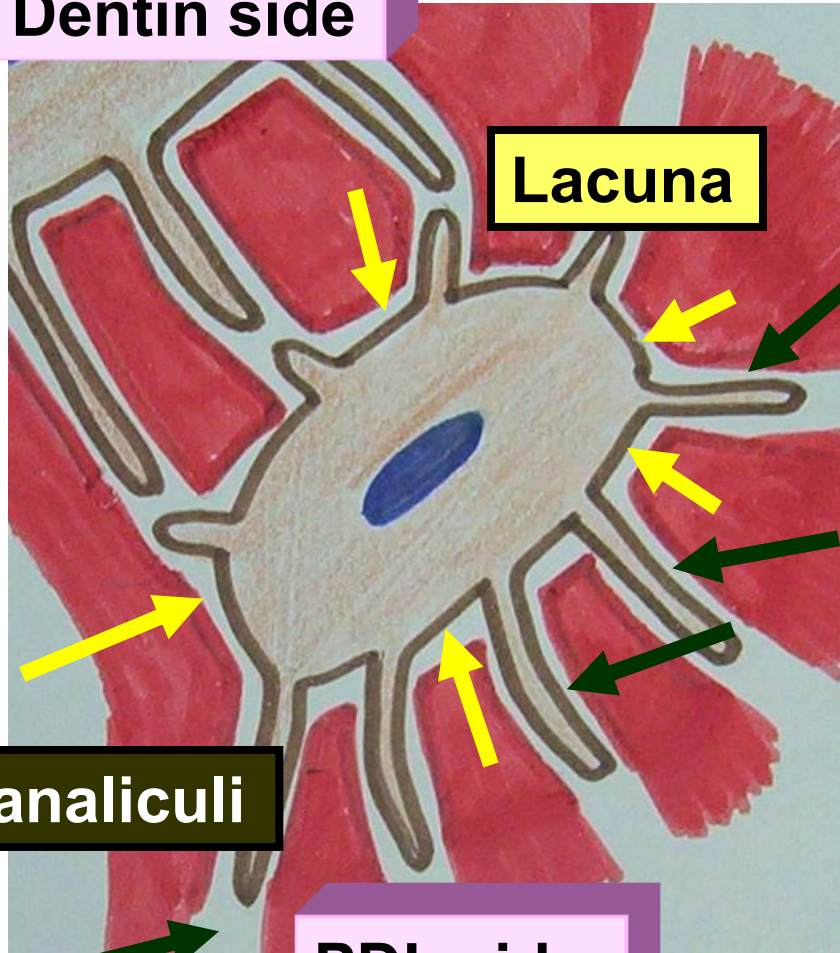
- Thickness is 150-200 μ .
- Covers the apical 1/3 of the root dentine.
- Similar to acellular cementum but contains cementocytes
- (imprisoned cementoblasts).
- Sharpey's fibers of periodontal ligament are inserted in it



Cementocytes

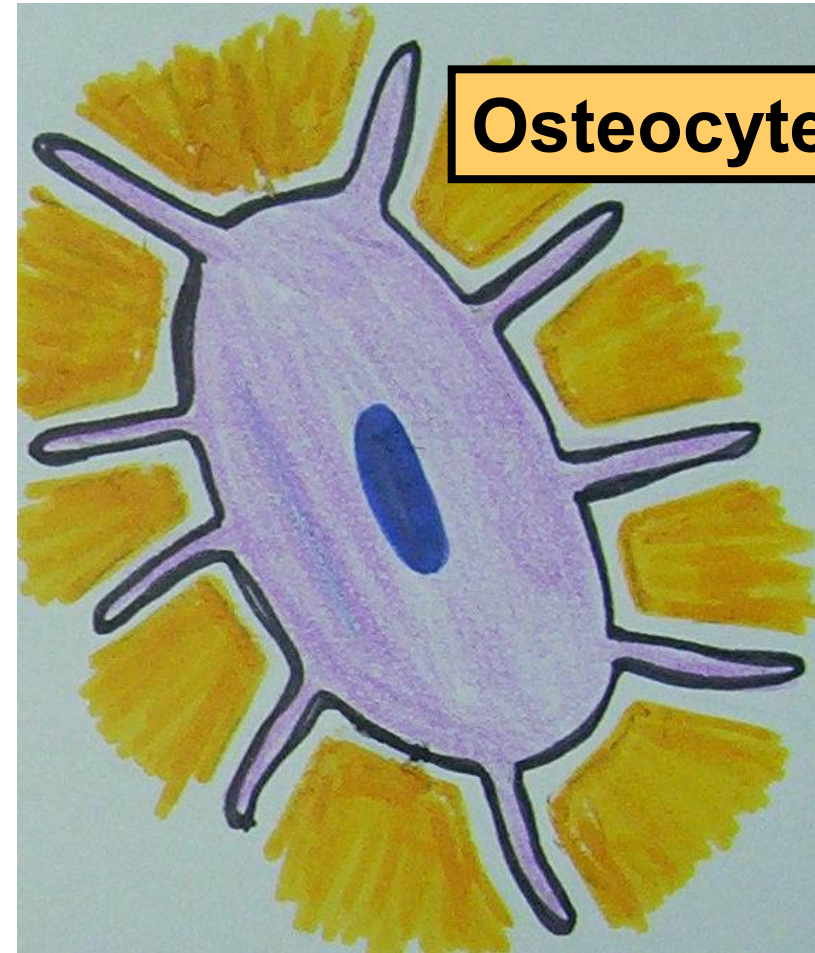
Cementocyte And ??????

Dentin side



PDL side

Osteocyte

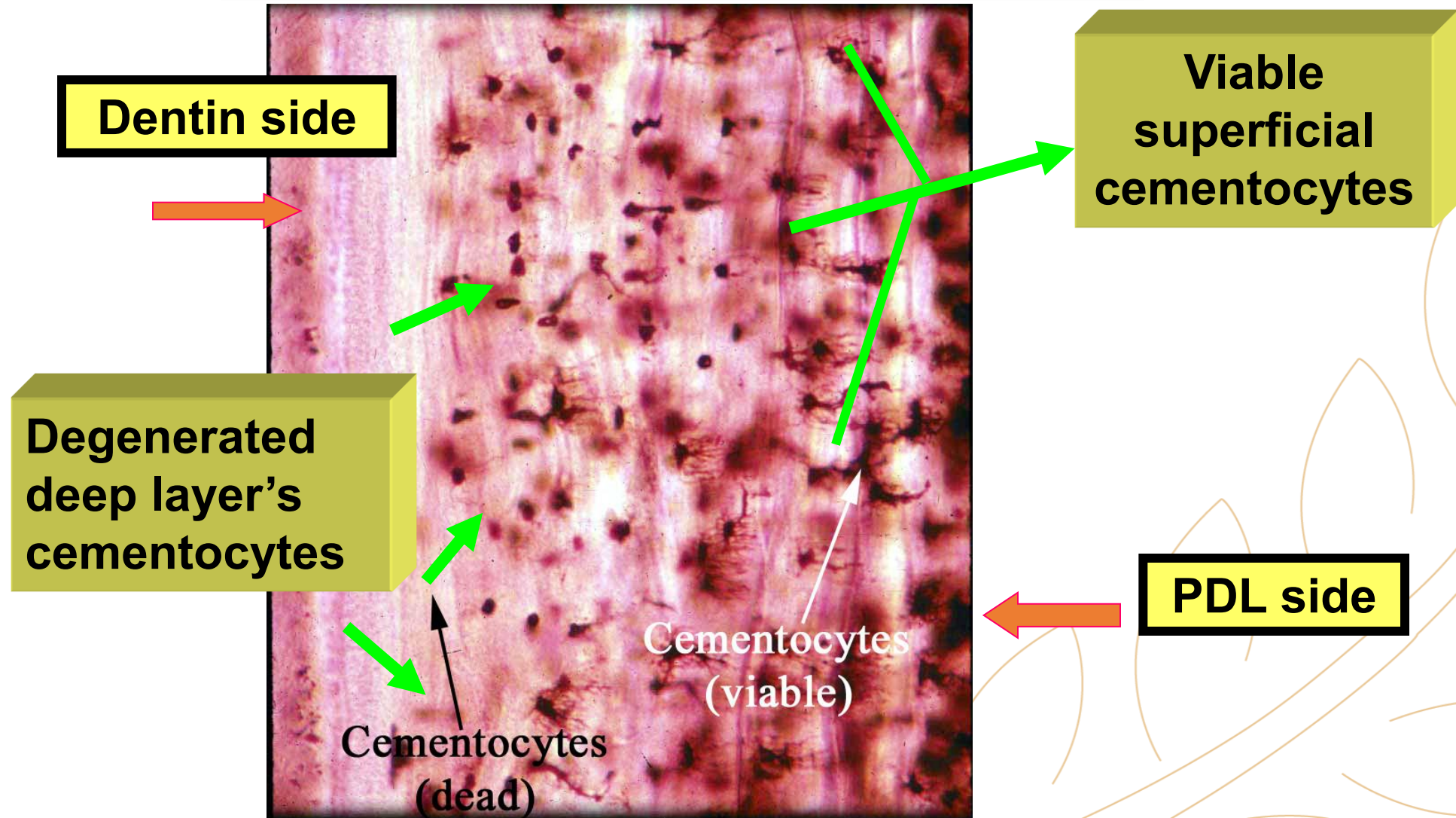


Cementocyte

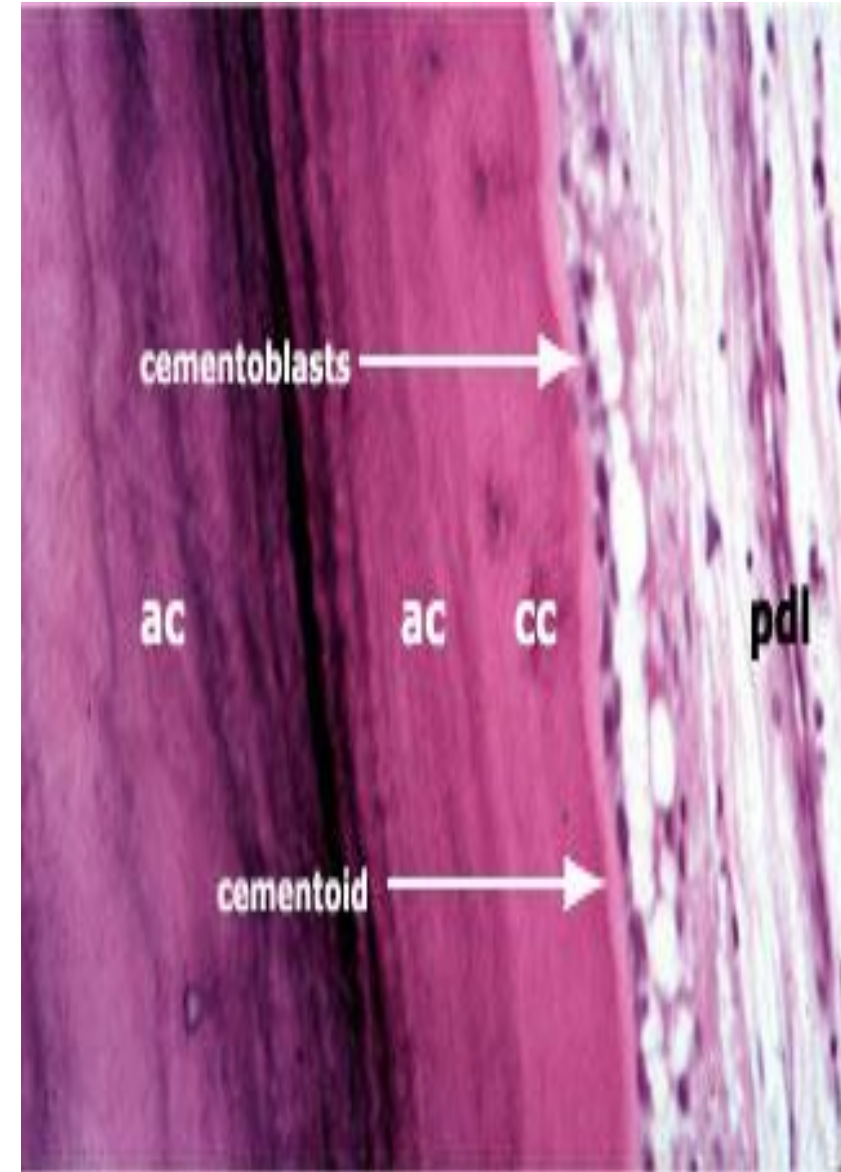
- Cell body is contained in a space called **lacuna**.
- Their processes are present in **canaliculi** and anastomose with those of the neighboring one.
- Most of the processes are directed to periodontal ligament side more than dentine side to provide nutrition for cementum.
- Cementocytes in the deepest layer show signs of degeneration.
- In ground section the lacunae appear black because they are burned leaving space filled with air.



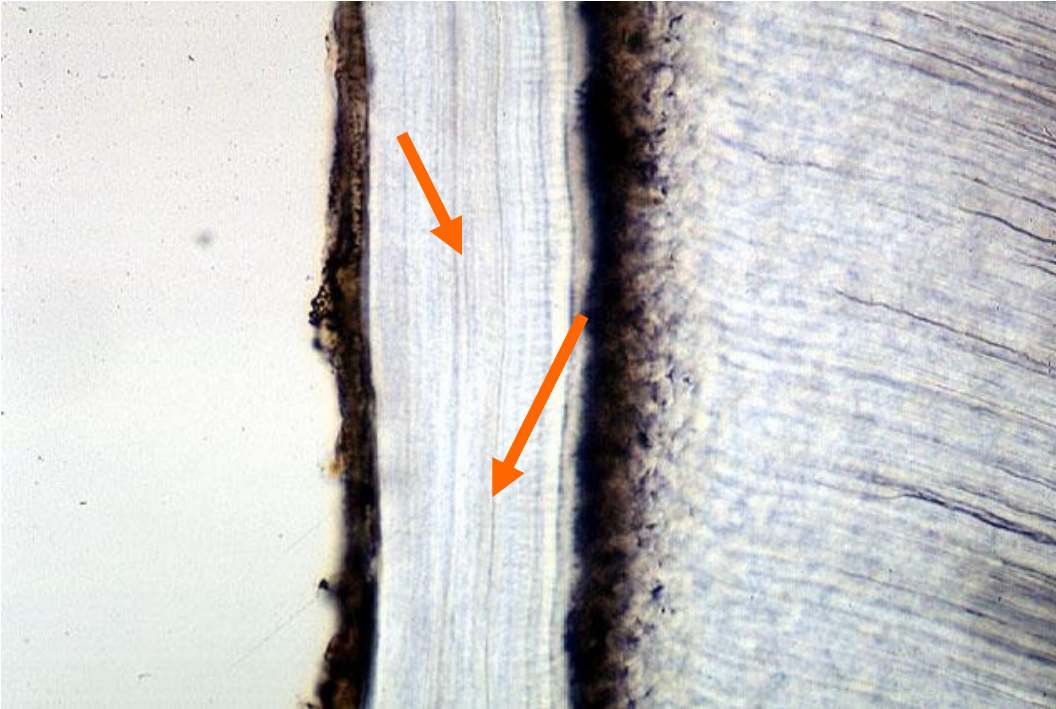
Cellular Cementum



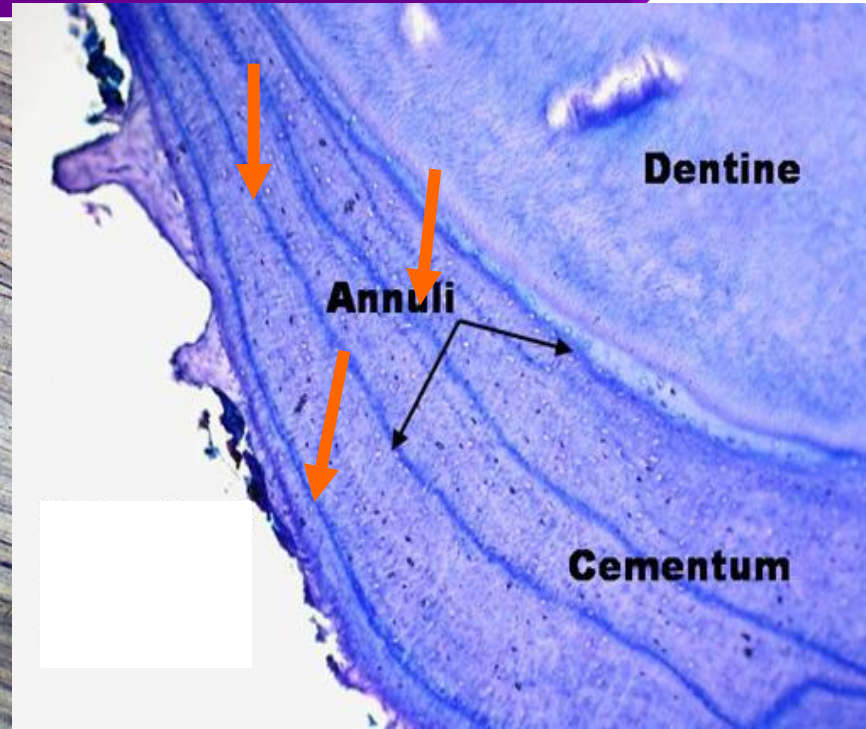
- Cementoblastic layer is always present on the outer surface of both cellular and acellular cementum resting on cementum matrix (cementoid layer).
- This denotes continuous formation of cementum throughout life .



Incremental Lines Of Salter



In acellular C



In cellular C

Incremental Lines Of Salter

- Due to rhythmic formation of cementum and present resting periods of cementoblasts.
- They run parallel to the long axis of the root .

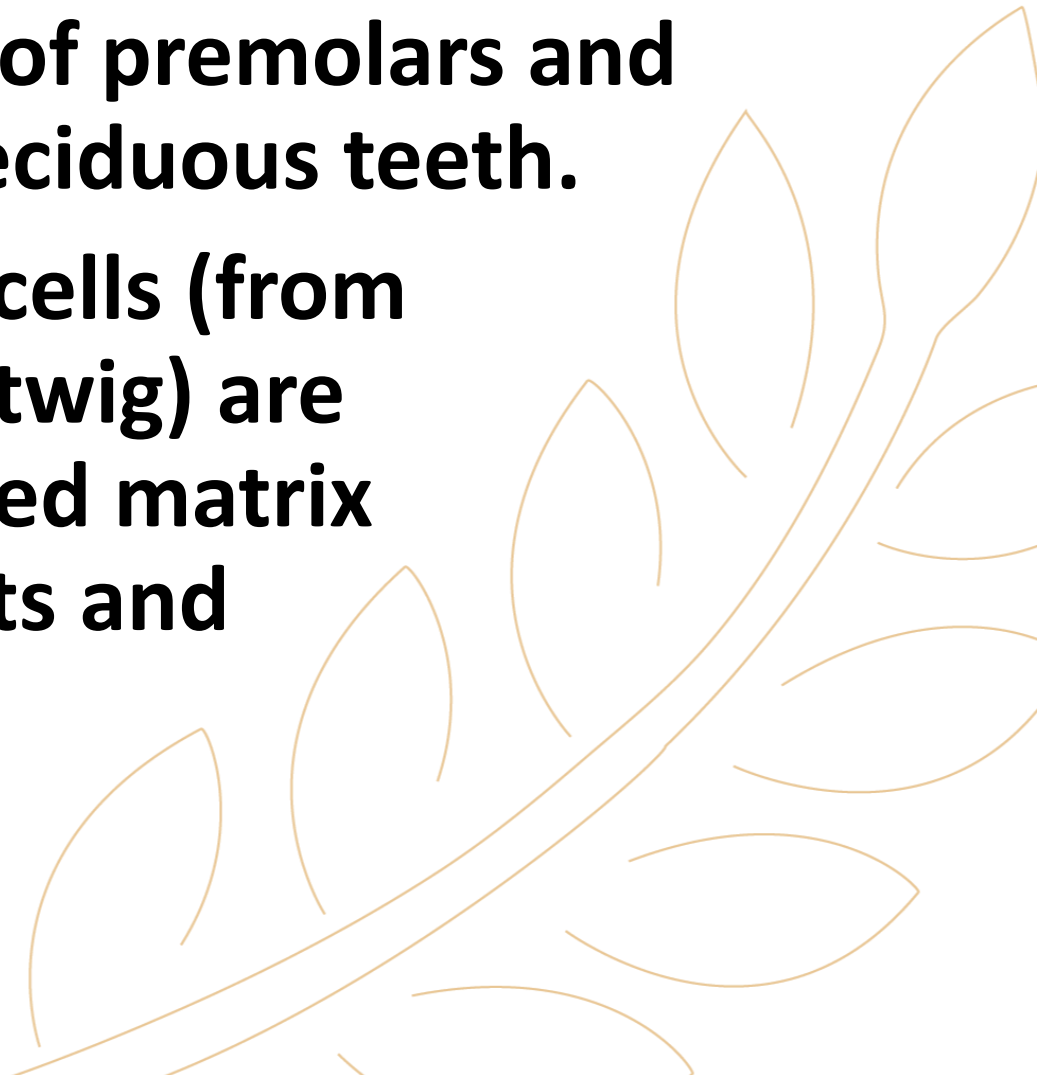
They are **hypermineralized** areas with less collagen fibers and more ground substance.

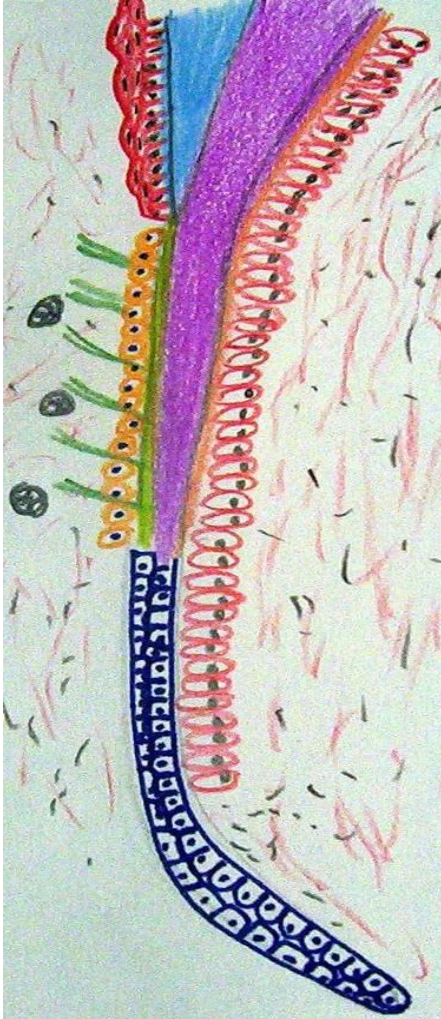
- The distance between incremental lines in cellular cementum is more than in acellular cementum.

What about neonatal line?

3-Intermediate Cementum

- **It is neither cementum nor dentine usually present in apical 1/3 of root of premolars and molars, rare in anterior or deciduous teeth.**
- **It represents a tissue where cells (from epithelial root sheath of Hertwig) are entrapped in rapidly deposited matrix (formed by both odontoblasts and cementoblasts).**

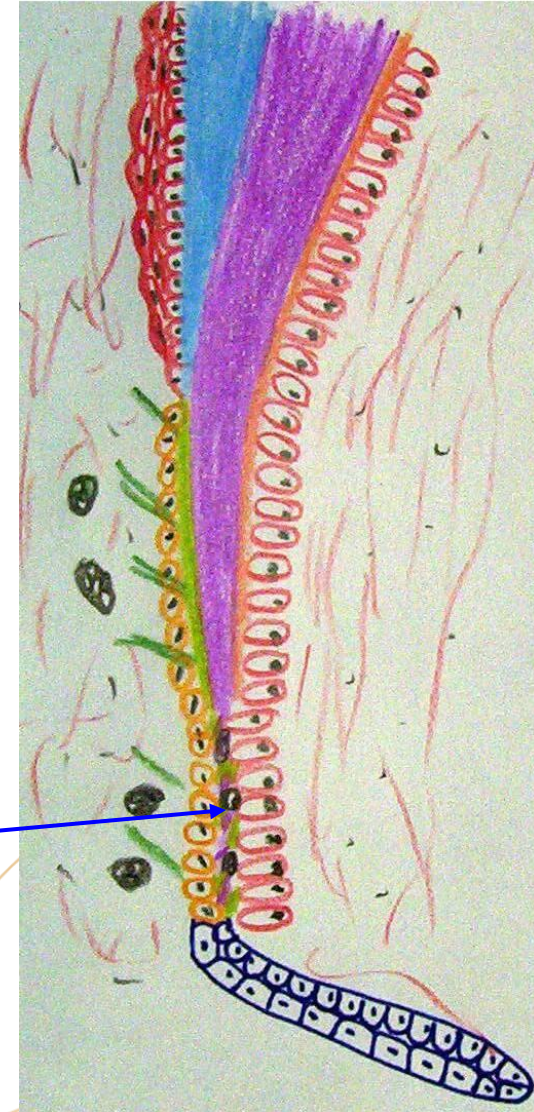


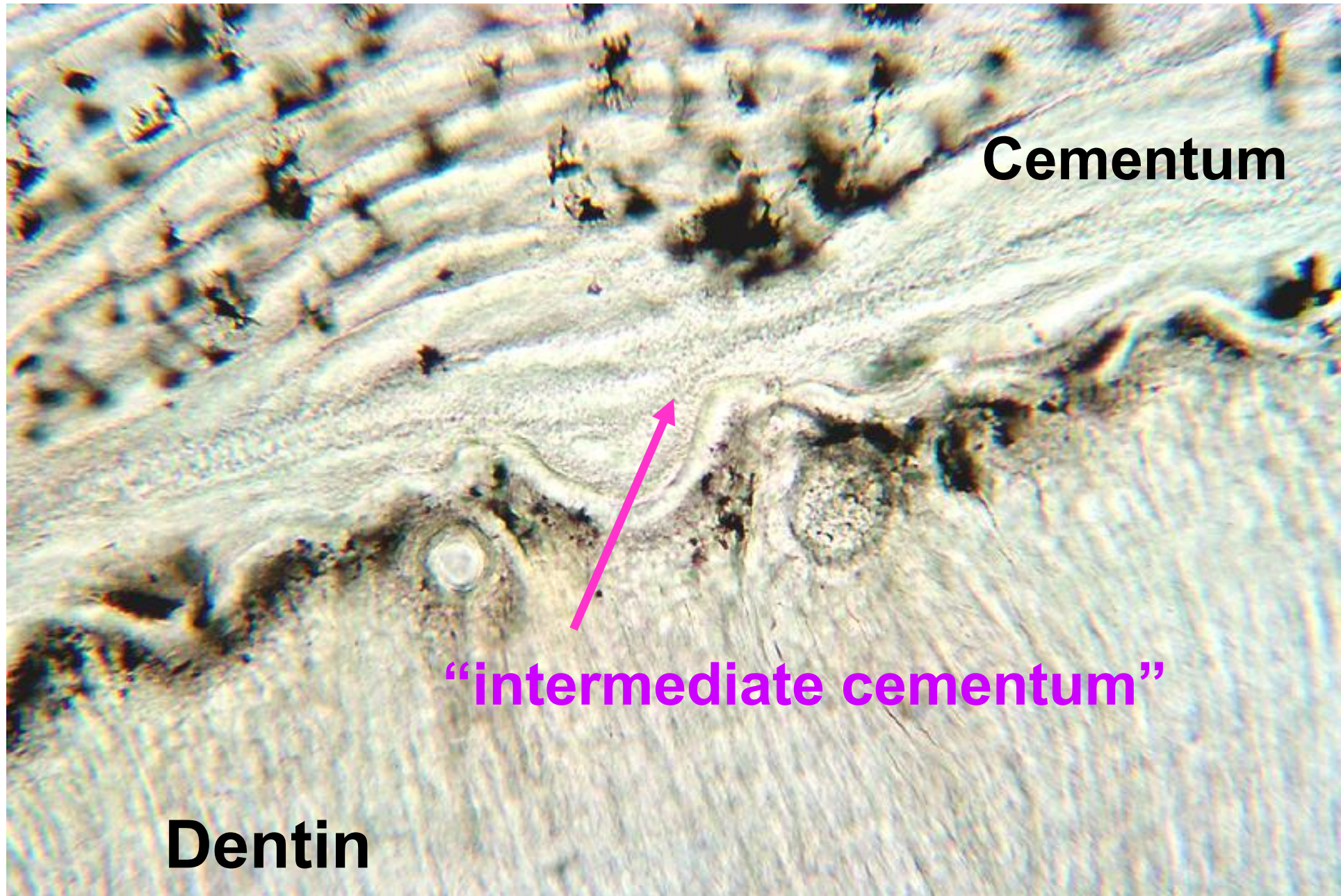


Cause:

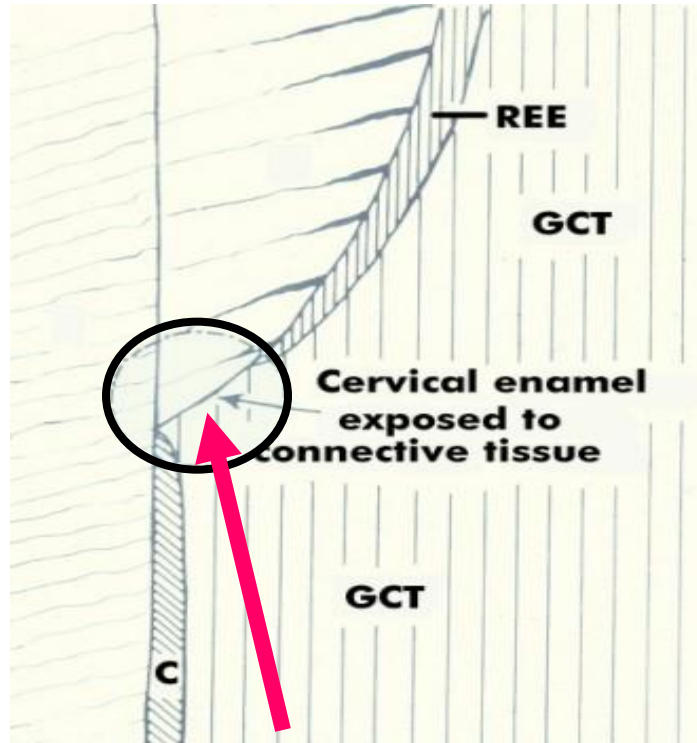
Premature degeneration of epithelial Root sheath of Hertwig after odontoblasts differentiation and before dentin formation.

Contains entrapped epithelial cells

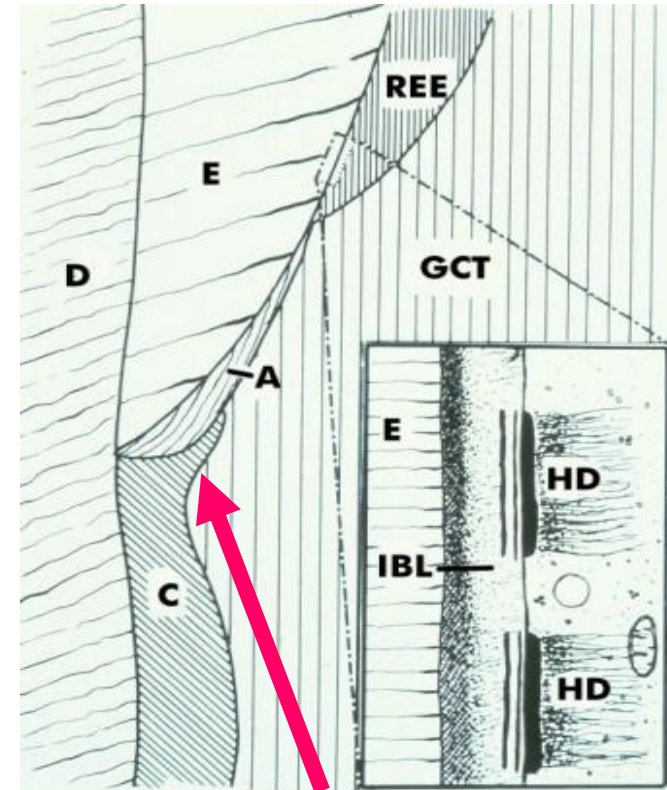




4-Afibrillar Cementum



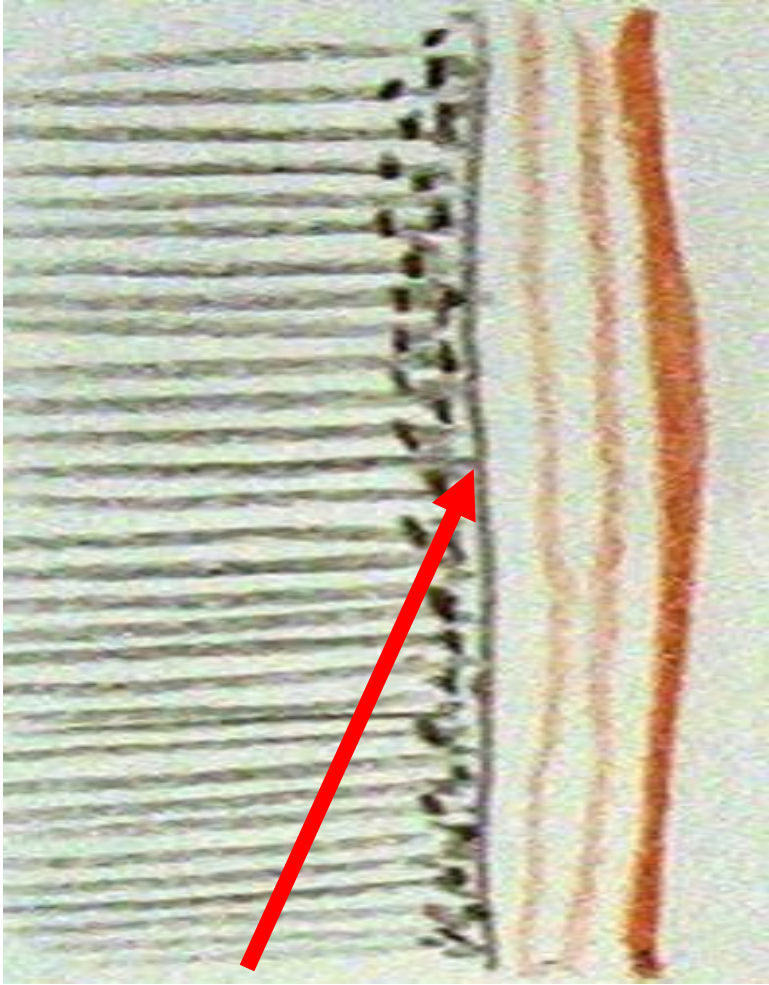
Early degeneration of the reduced dental epithelium before tooth eruption results in the enamel at cervical area is not protected by it.



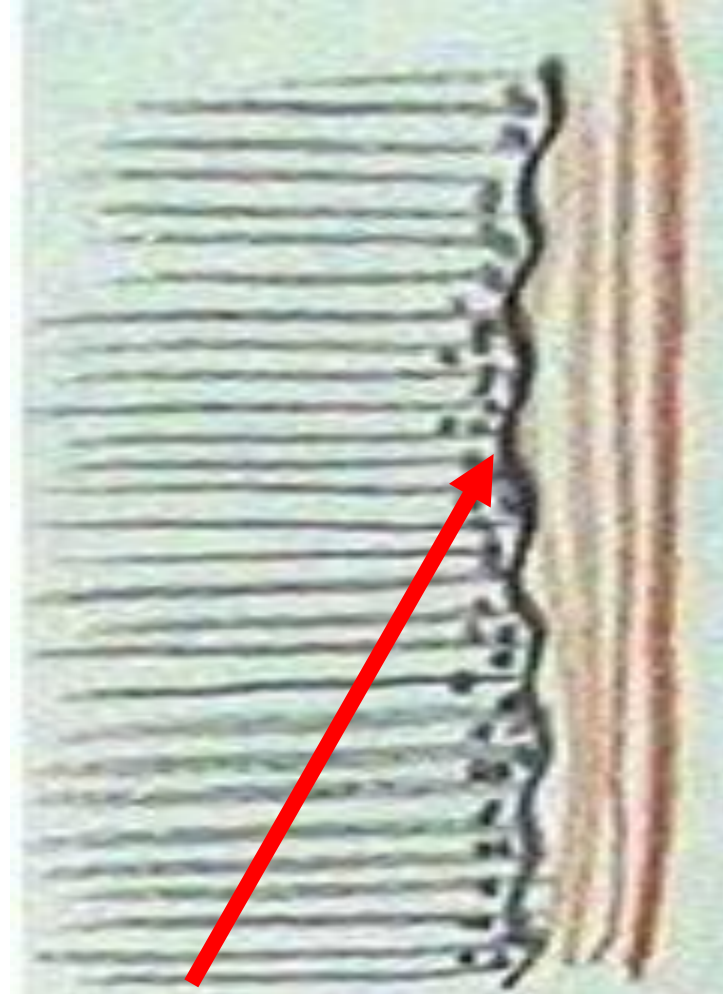
The connective tissue of the dental sac lay down cementum on the exposed enamel

It is a thin layer of well mineralized ground substance

Cemento Dentinal Junction



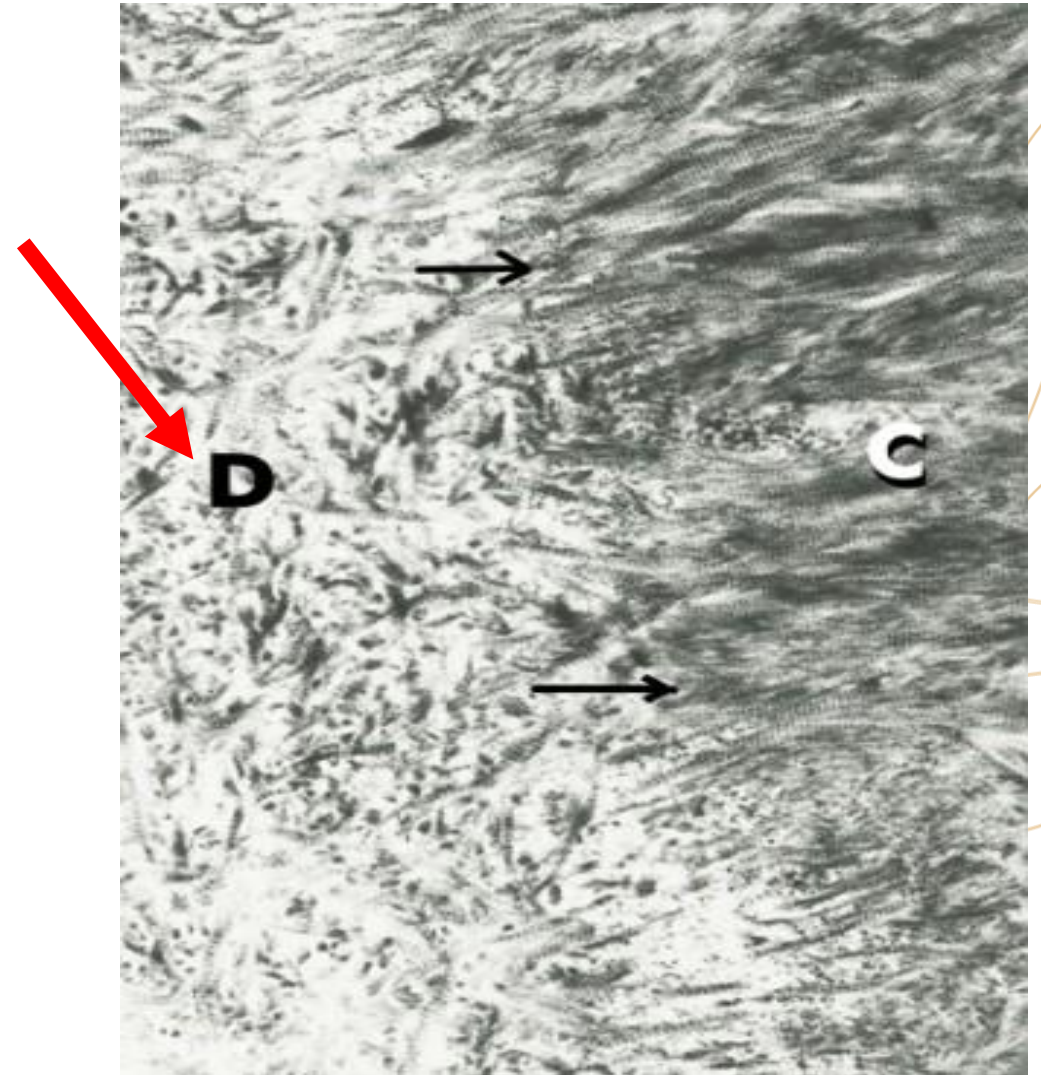
Smooth in permanent teeth



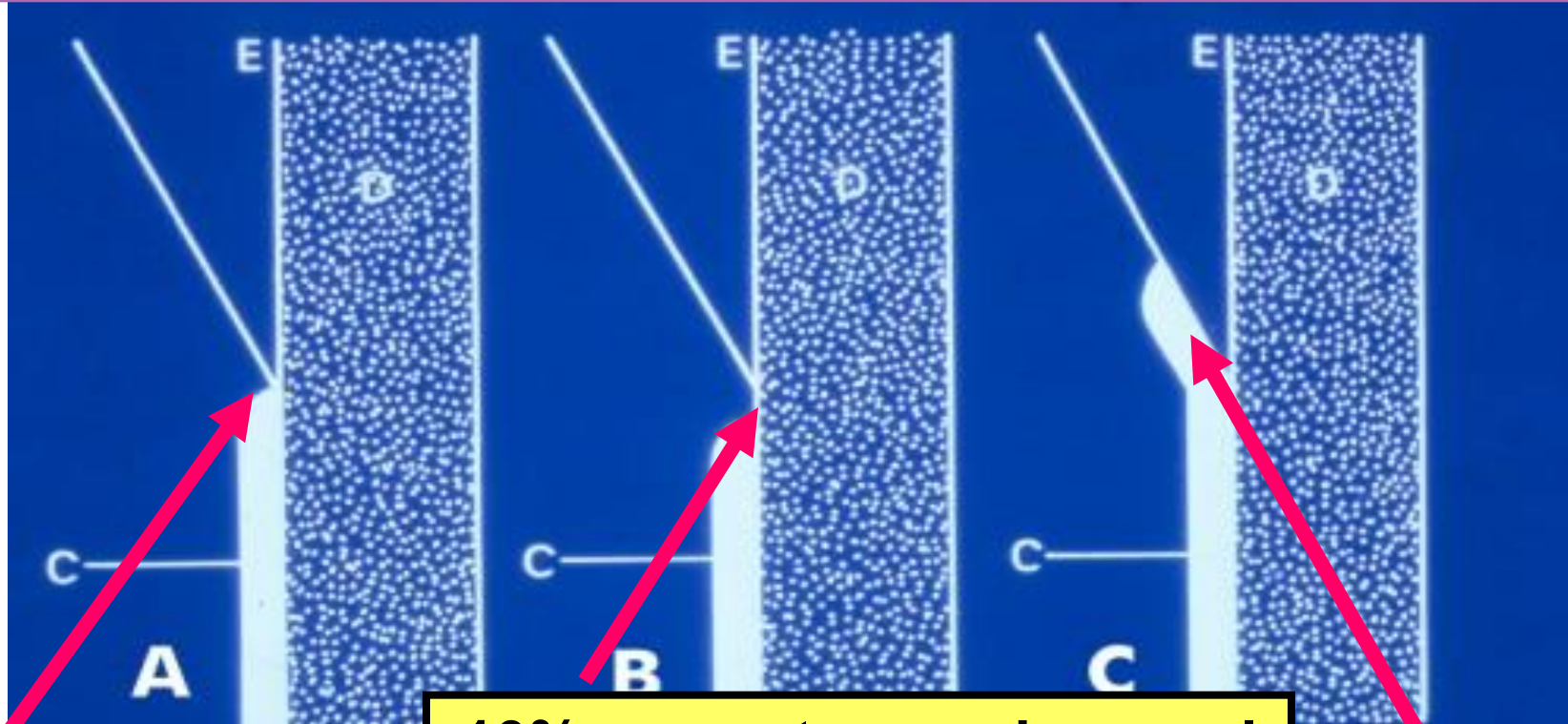
Scalloped in deciduous teeth

Cemento Dentine Junction

- Clearly seen in decalcified sections
- But,
- When viewed by EM the junction is not seen clearly due to intertwining of collagen fibers of cementum and dentin.



Cemento Enamel Junction



30% cementum meets the enamel in a sharp line

10% cementum and enamel doesn't meet because of delayed separation of epith root sheath of Hertwig (area of dentin not covered by C).

60% cementum overlaps E (afibrillar cementum)

Cementogenesis

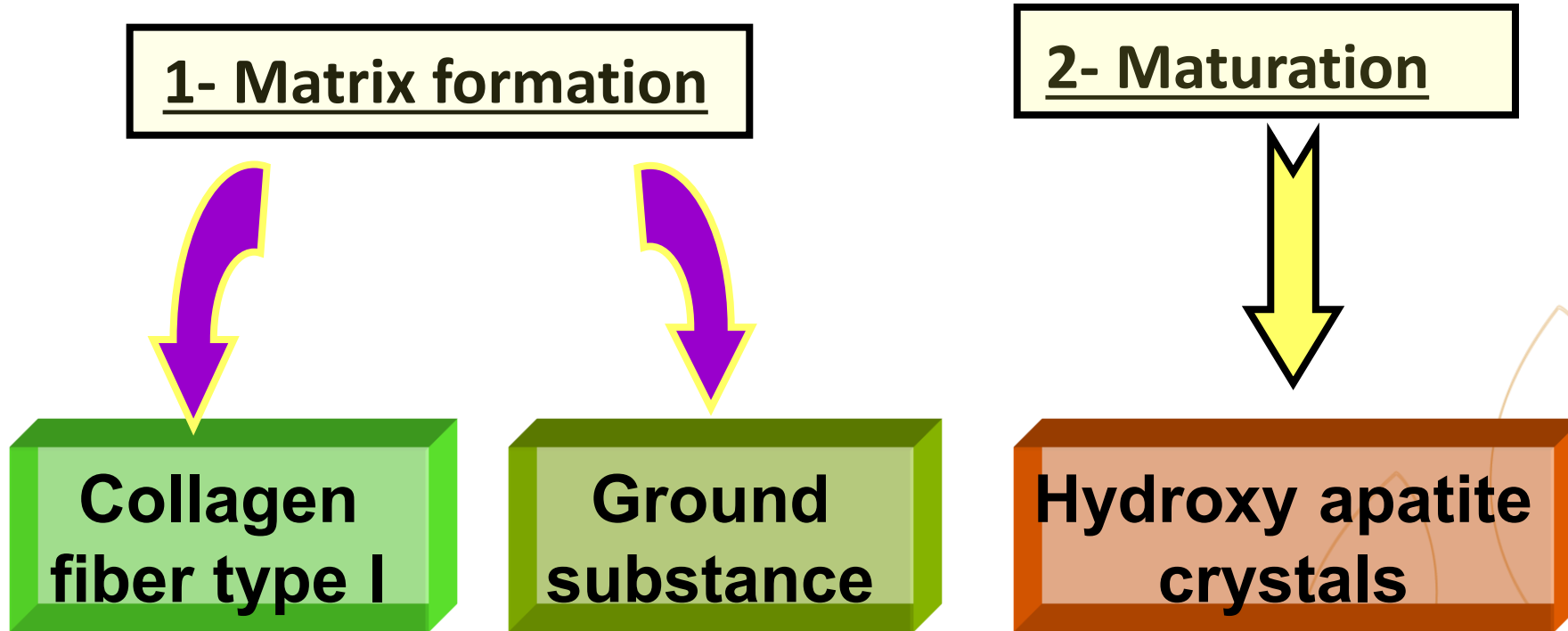
1- Matrix formation

**Collagen
fiber type I**

**Ground
substance**

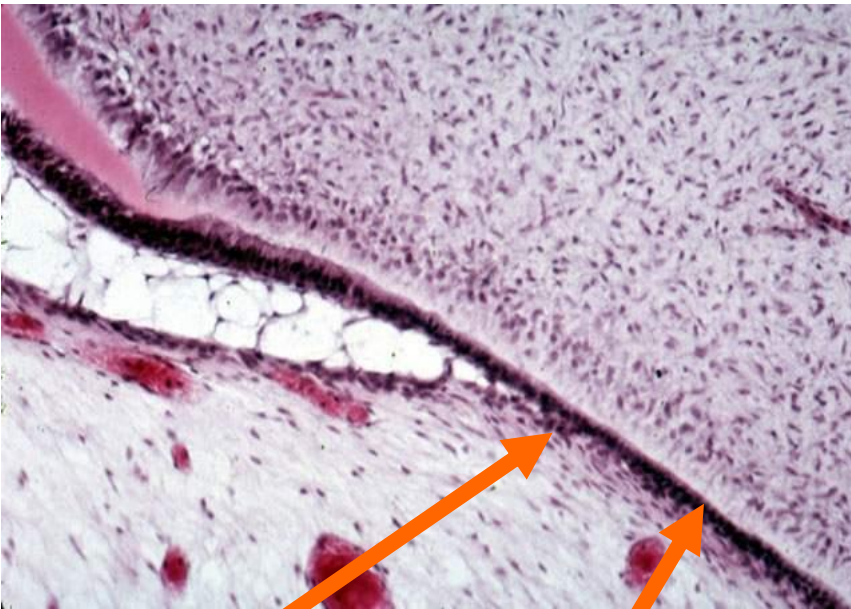
2- Maturation

**Hydroxy apatite
crystals**



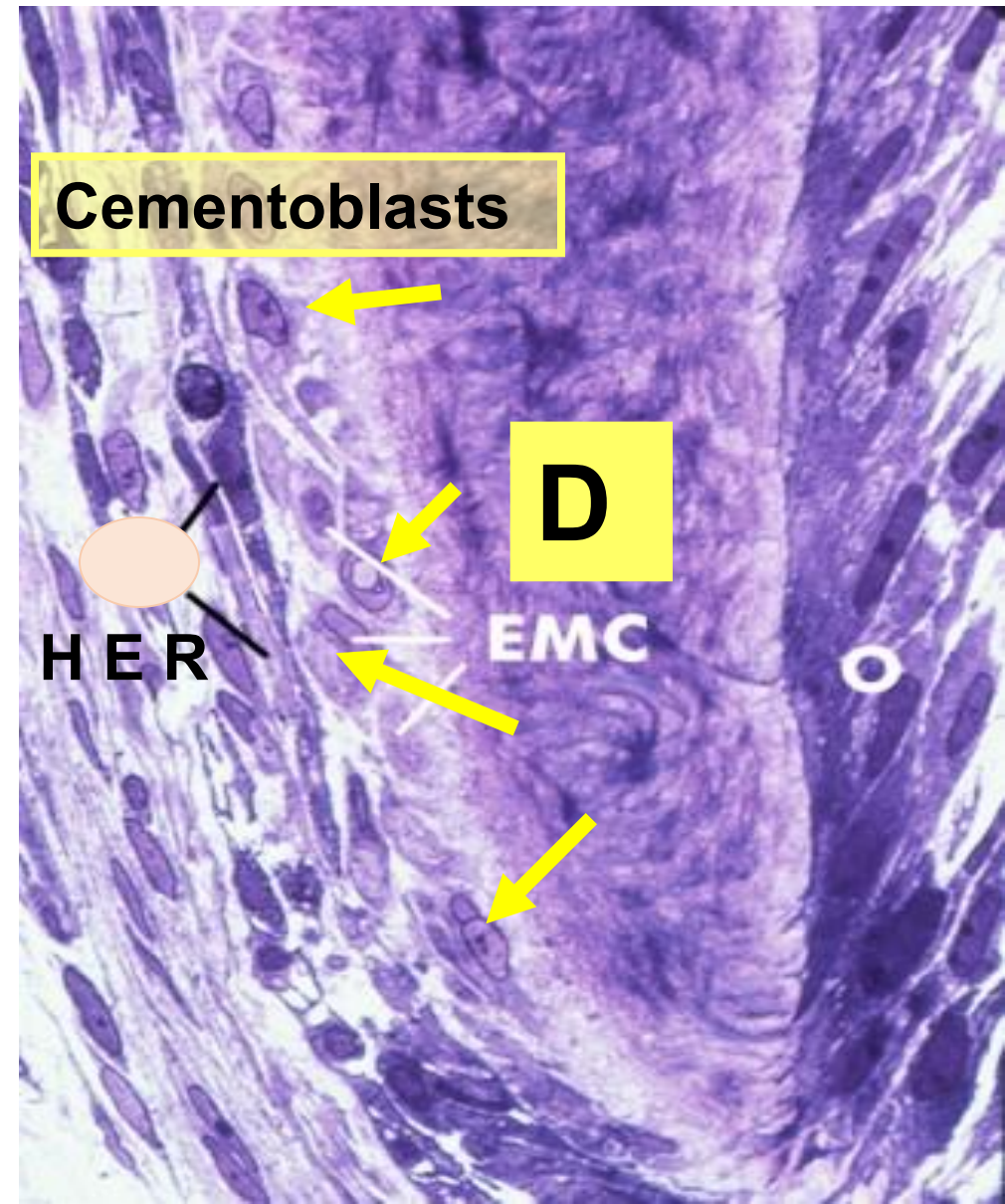
1- Matrix formation

Cementum matrix (collagen F, ground substance) is formed by the cementoblasts during root formation



Future C E J

Epith. Diaph.



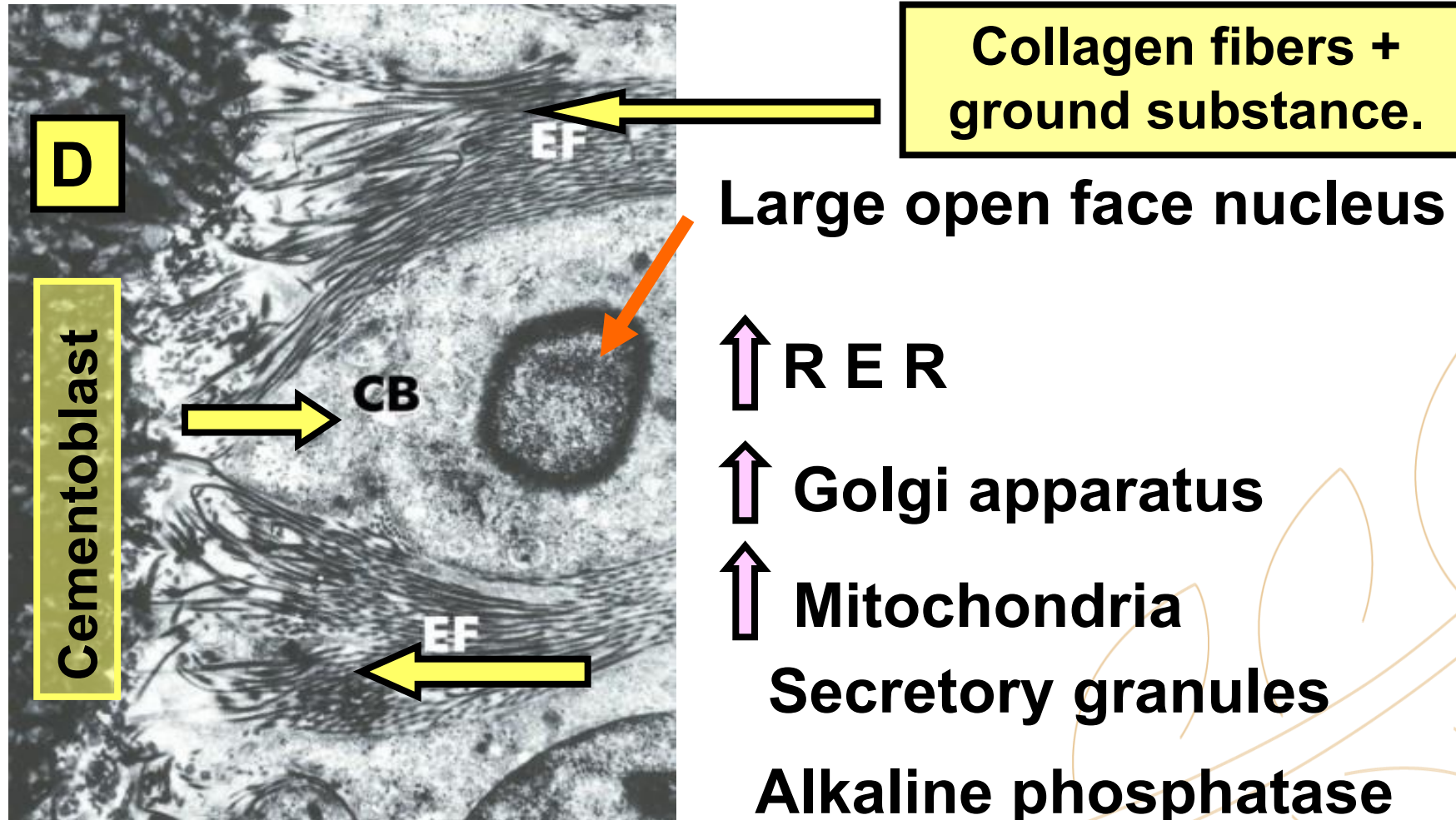
1- Matrix formation

- As in root formation, when enamel and dentine reach C.E.J.
- Inner and outer enamel epithelium of cervical loop will form epithelial root sheath of Hertwig (the bended part is called epithelial diaphragm).
- I.E.E. of epithelial root sheath of Hertwig causes differentiation of odontoblasts which will form dentine. Once dentine is formed, the Hertwig will degenerate (the remnants is epithelial rests of Malassez).
- U.M.Cs. of dental sac differentiated into cementoblasts which have the characters of protein forming and secreting cells.
- Cementoblasts produce the matrix (collagen & ground substance).

1- Matrix formation

Cementoblast is a protein forming and secreting cell.

The active cementoblasts are rounded.



2-Maturation:

Maturation occur layer by layer where Ca Ph hydroxyapatite crystals will be deposited in between and on the surface of collagen fibers having parallel arrangement with collagen fibers.

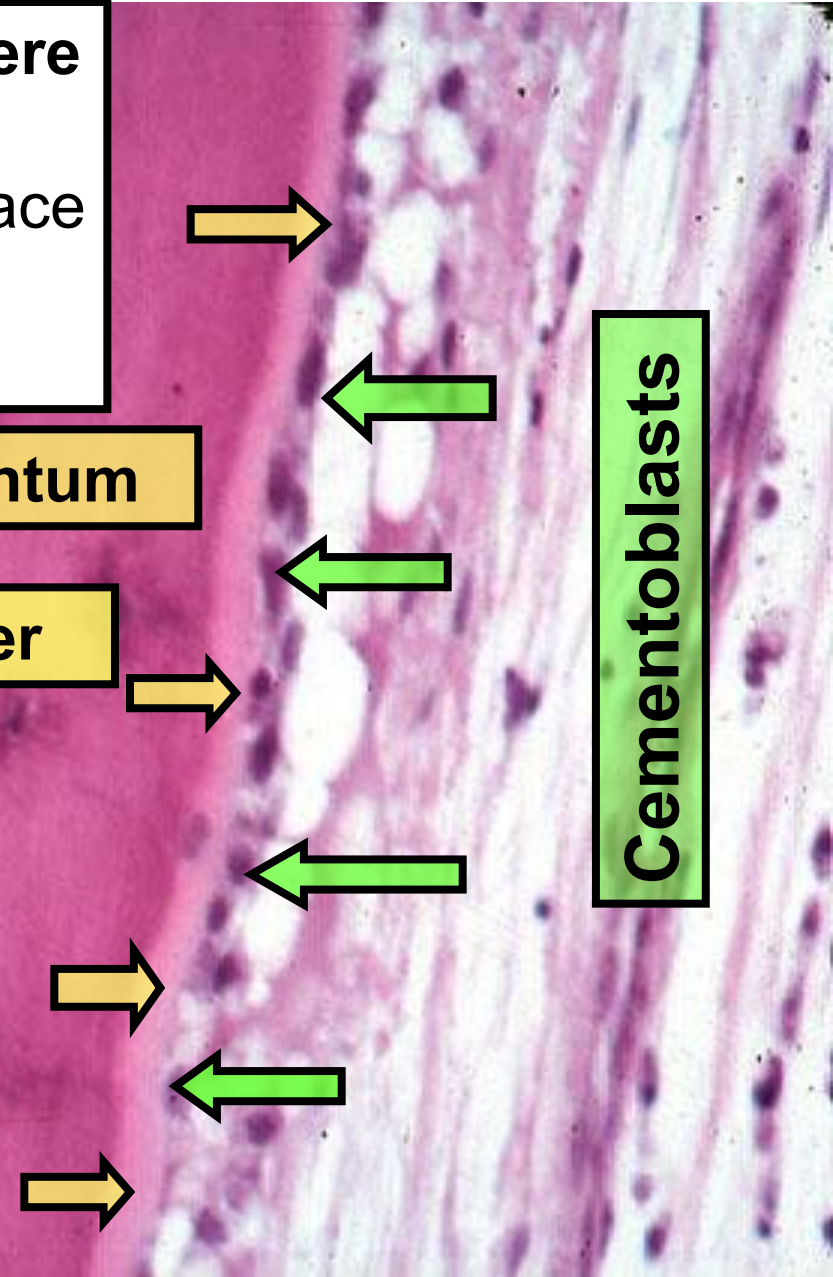
Cementum

Cementoid layer

Cementoblasts

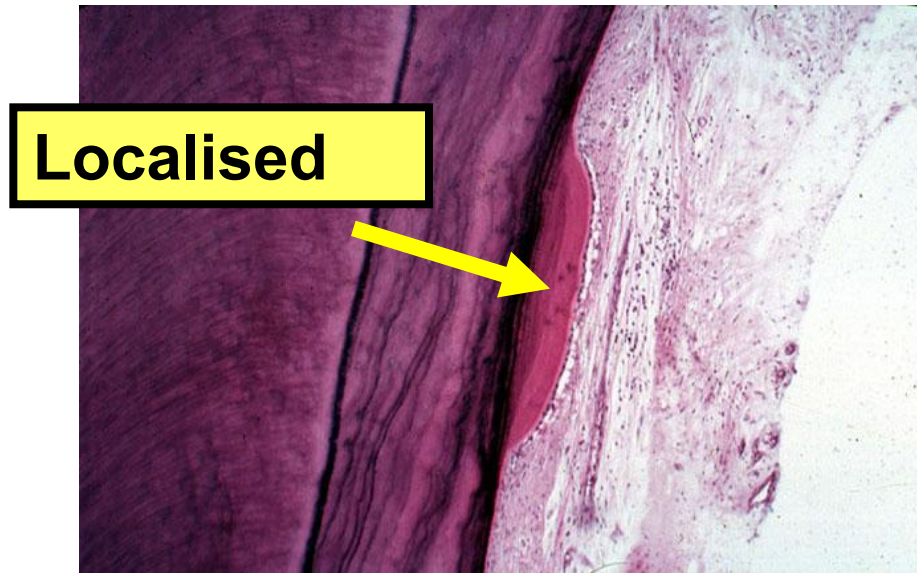
N.B.

The outer surface of the root (wether in cellular or a cellular cementum) is covered by cementoid tissue lined by cementoblasts.

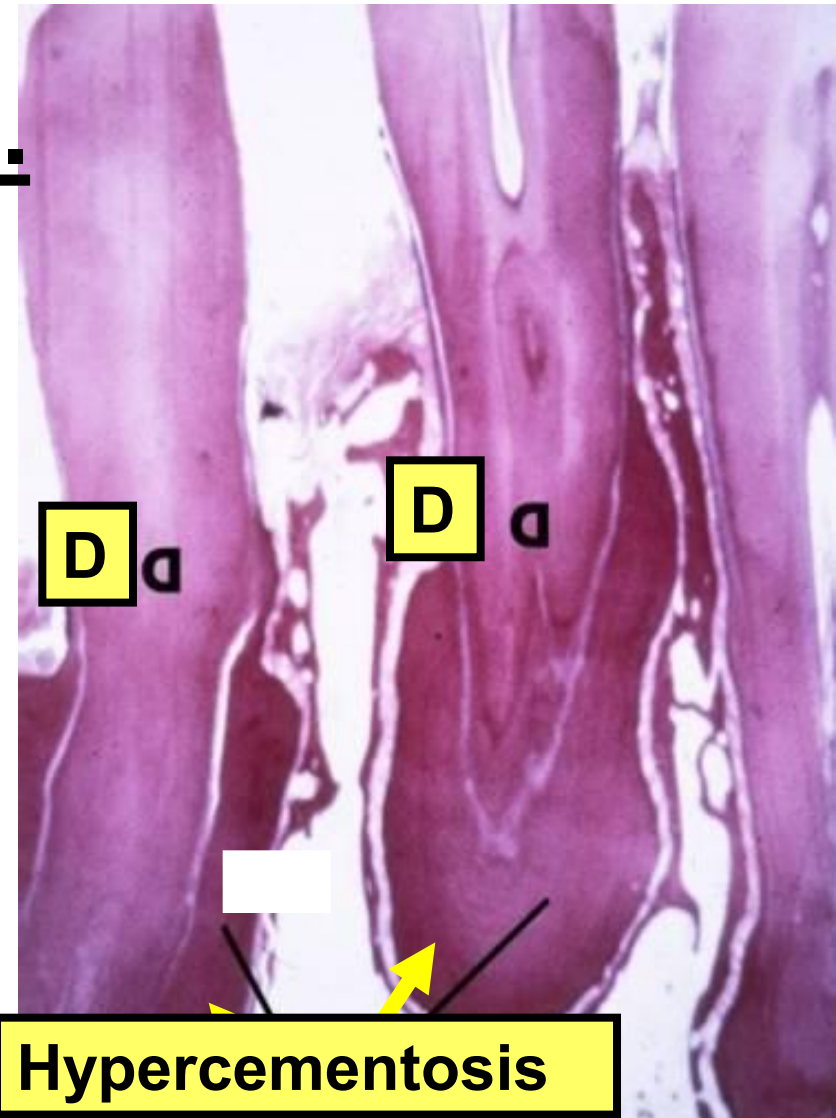


Age Changes Of The Cementum

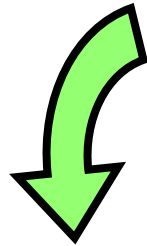
1- Hypercementosis.



May affect one tooth (all the tooth or part of it)or all teeth



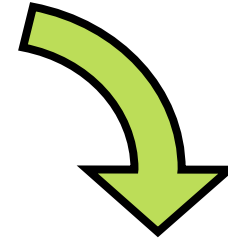
Types Of Hypercementosis



**Hypercementosis
hypertrophy**



**Increase number of
Sharpey's fibers in
functioning teeth**



**Hypercementosis
hyperplasia**



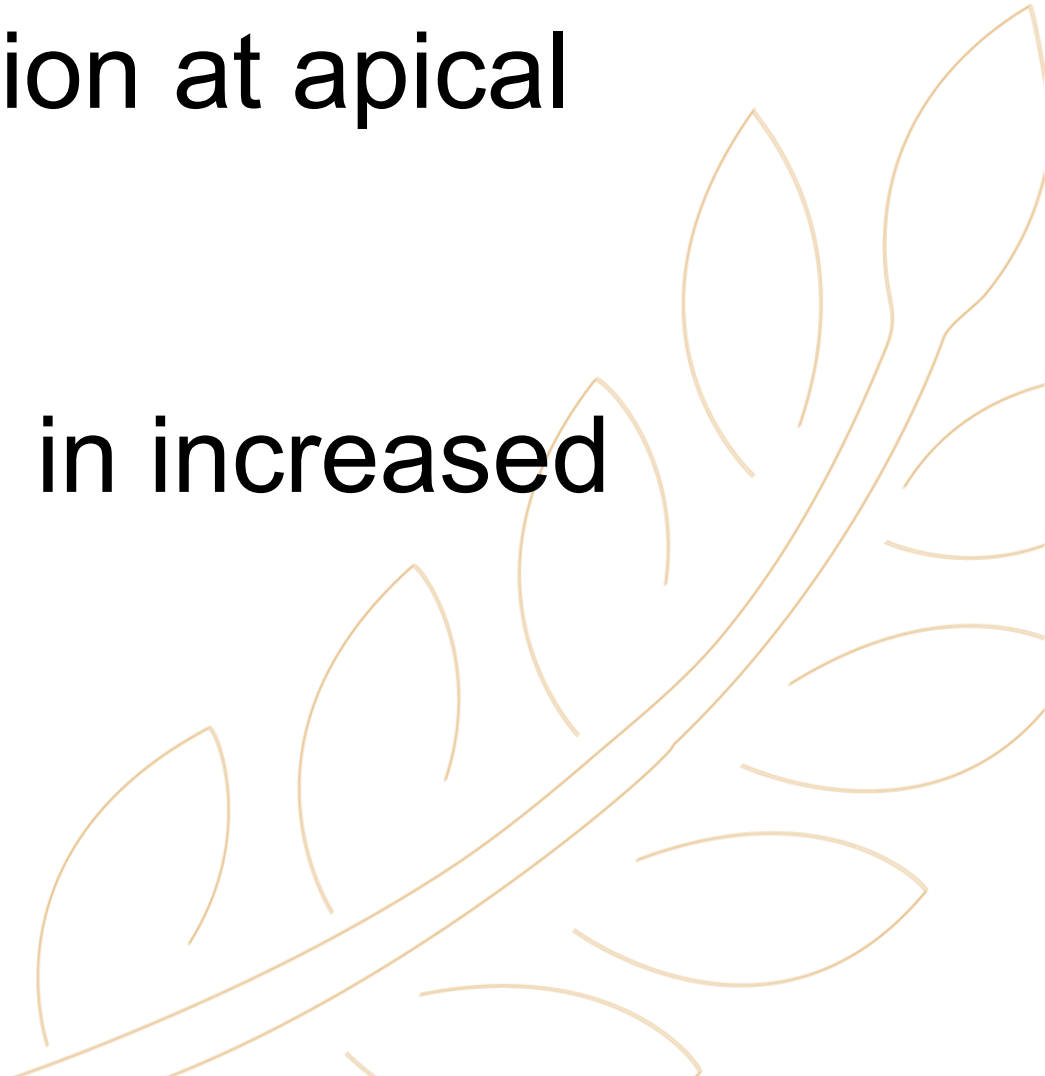
**Decrease number of
Sharpey's fibers in
non-functioning teeth**

1- Cemental Hypertrophy:

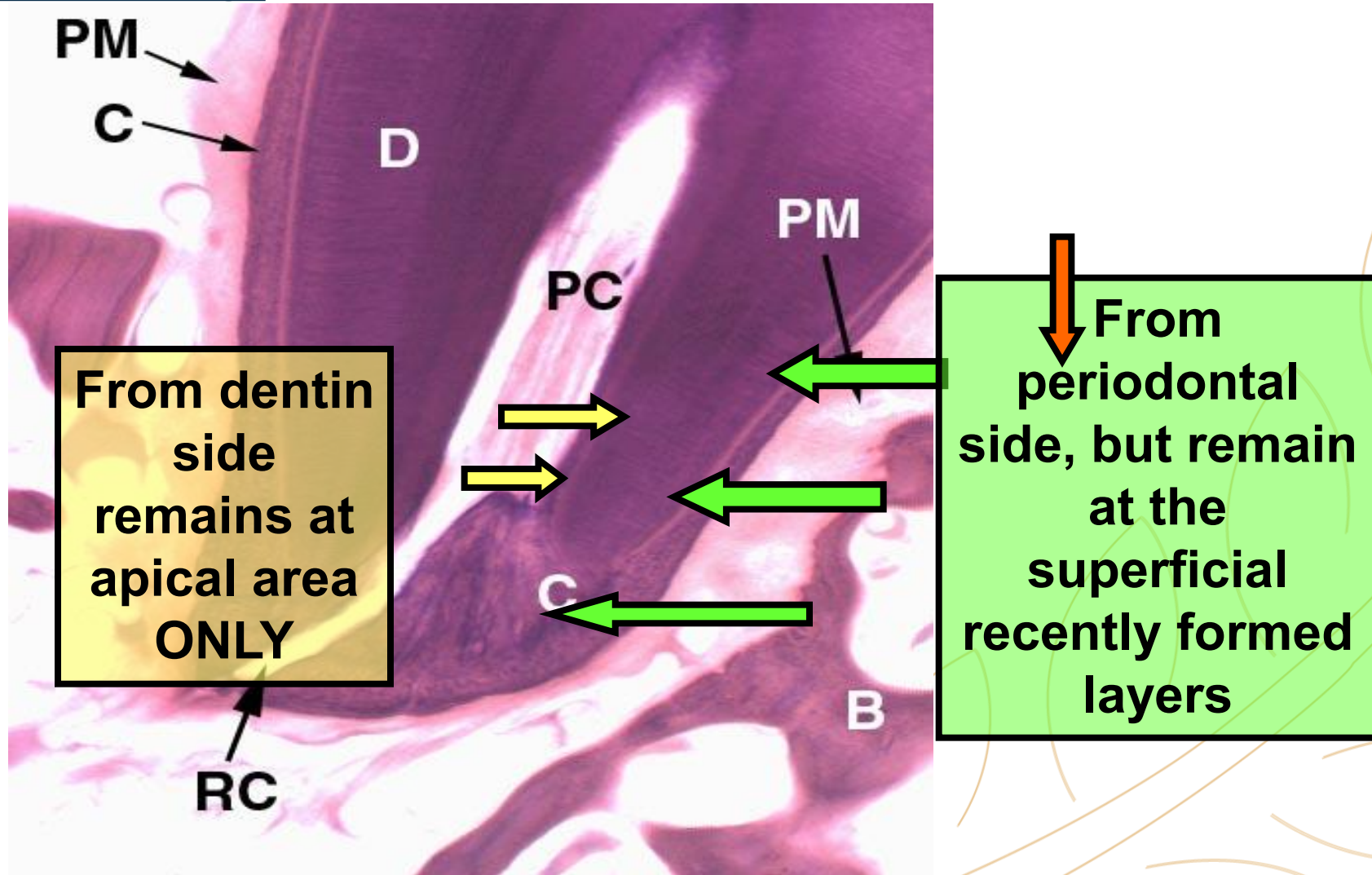
Increased cementum formation at apical third (functional adjustment)

2- Cemental Hyperplasia:

Defective condition resulting in increased cemental formation.

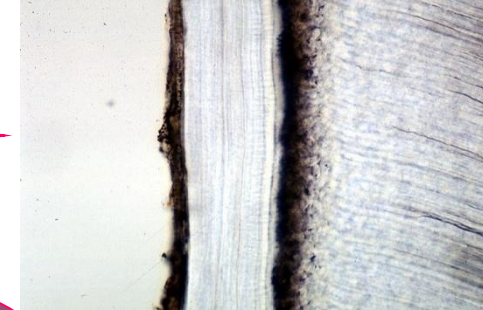


2- Permeability



Types Of Cementum

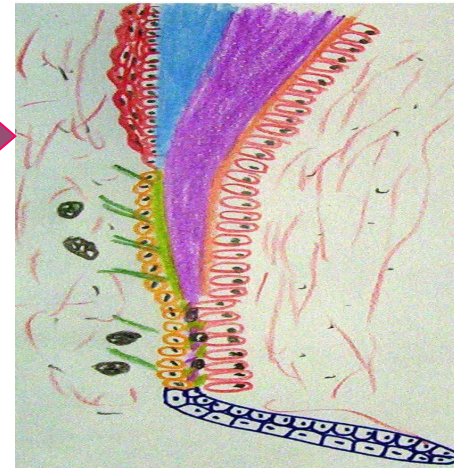
- 1- Acellular cementum



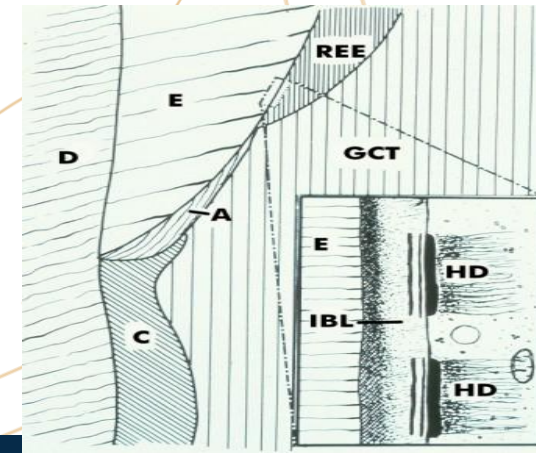
- 2- Cellular cementum



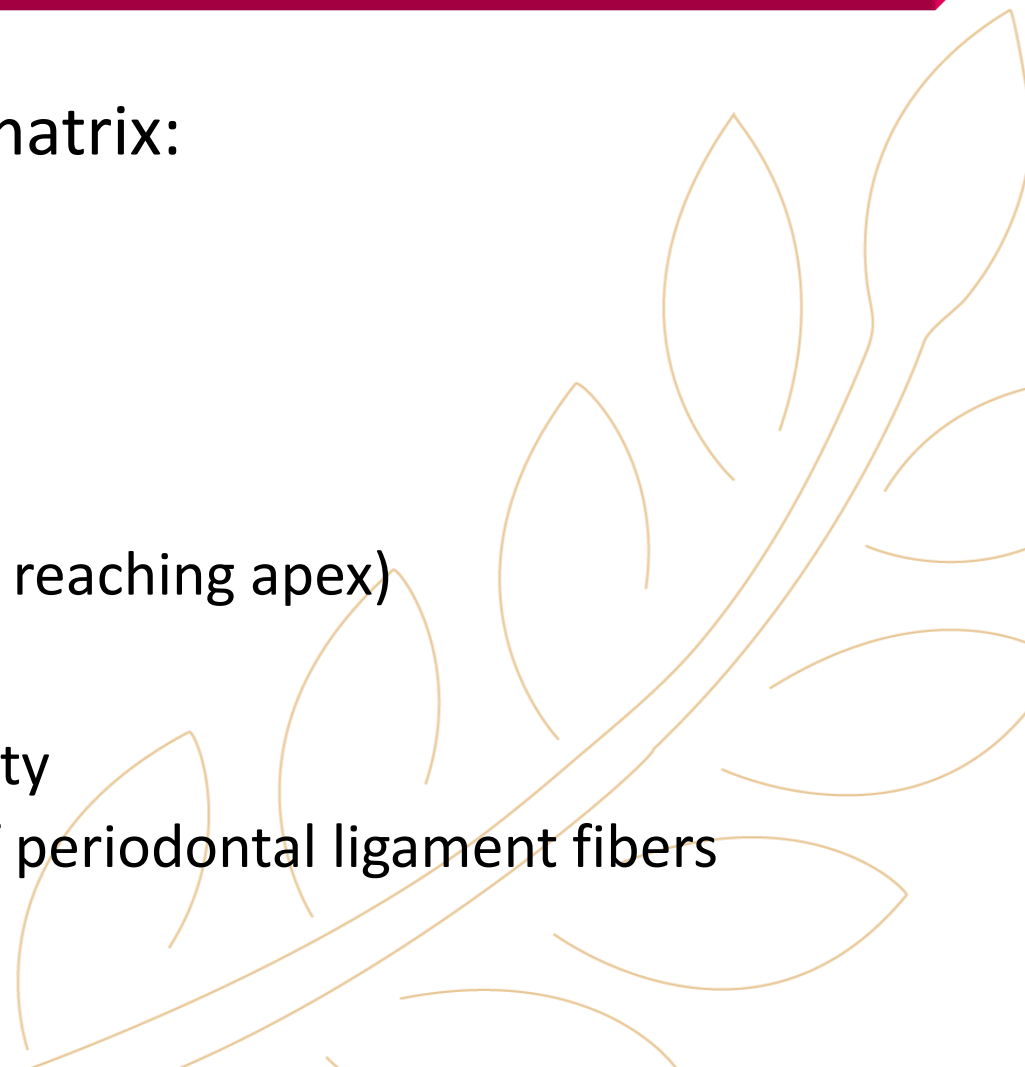
- 3- Intermediate cementum



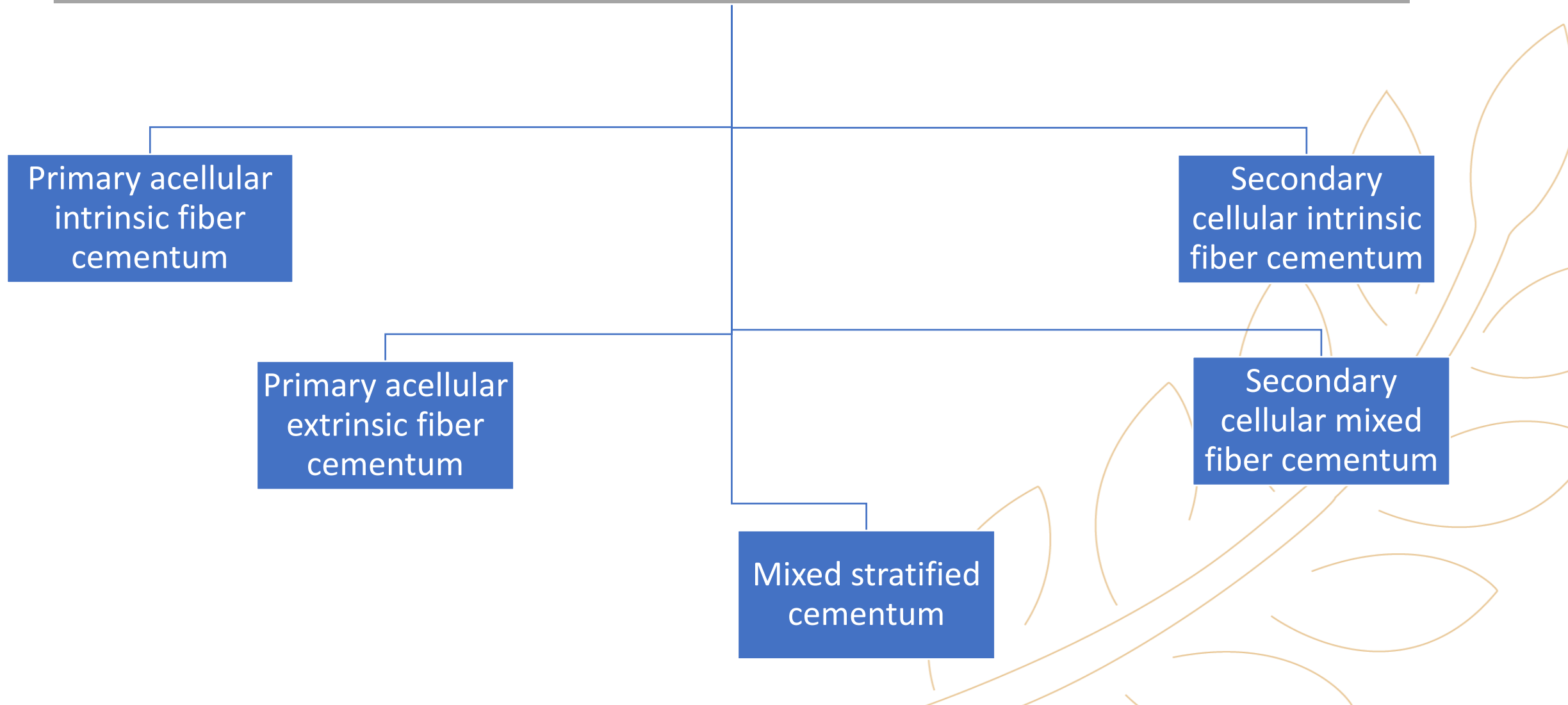
- 4- Afibrillar cementum



Cementum could be classification according to:

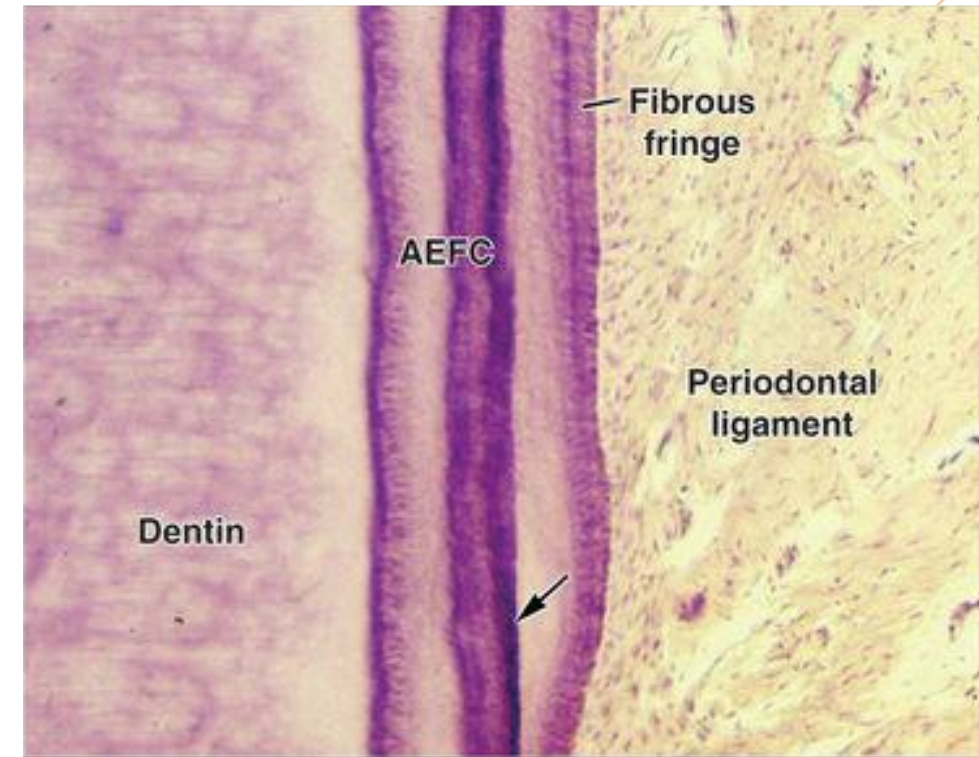
- A. The presence or absence of cells within its matrix:
 - Acellular cementum
 - Cellular cementum
 - B. Time of formation
 - Primary cementum
 - Secondary cementum (after cementum formation reaching apex)
 - C. The origin of collagen fibers in the matrix
 - Intrinsic fibers resulting from cementoblasts activity
 - Extrinsic fibers resulting from the incorporation of periodontal ligament fibers
- 

Accordingly, the following types are described



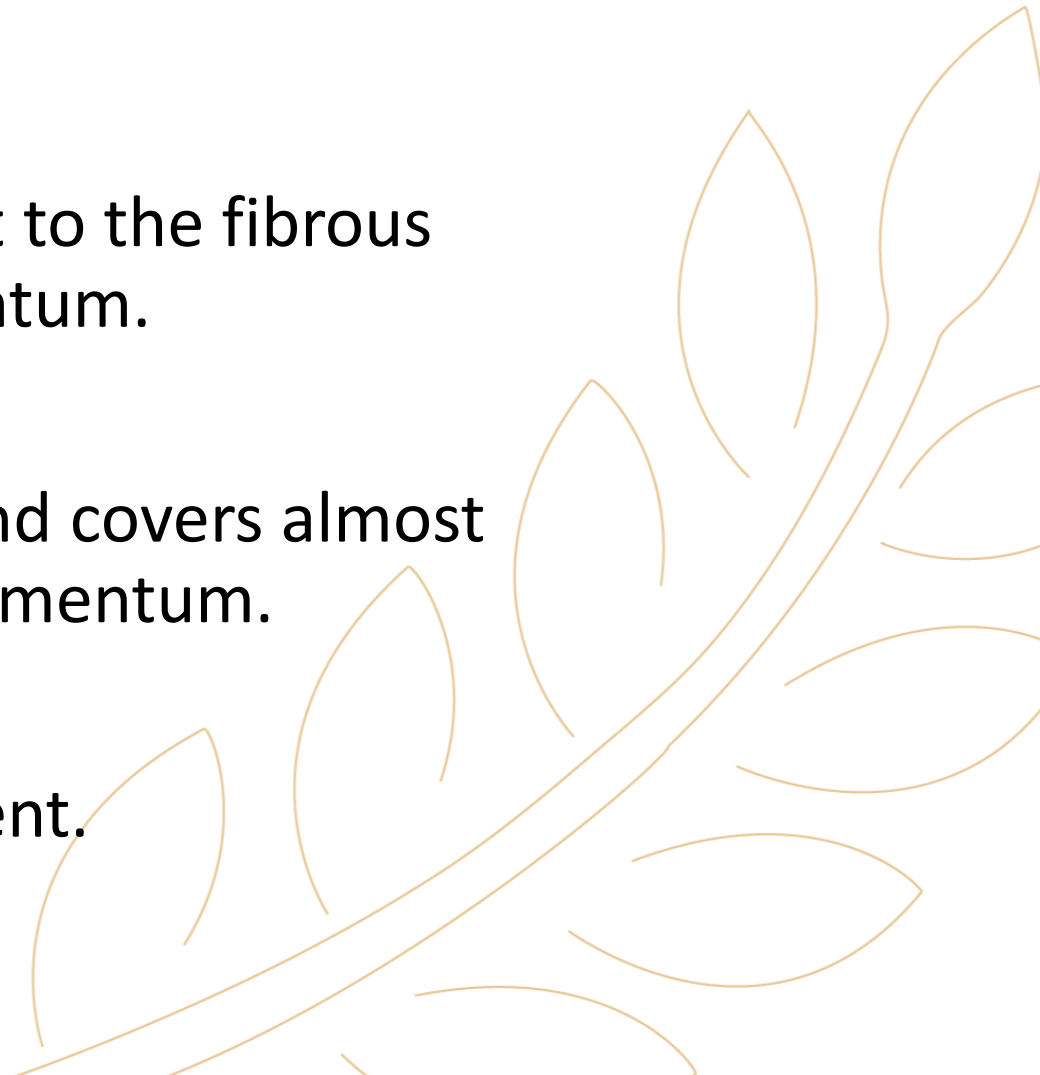
Primary acellular intrinsic fiber cementum

- Forms 10% of acellular cementum
- Matrix: Is the result of cementoblast (completely) activity and is elaborated before formation of PDL so all collagen is intrinsic.
- Thickness and site: innermost 15-20 μ m of cementum
- Function: Attachment of cementum & dentin together

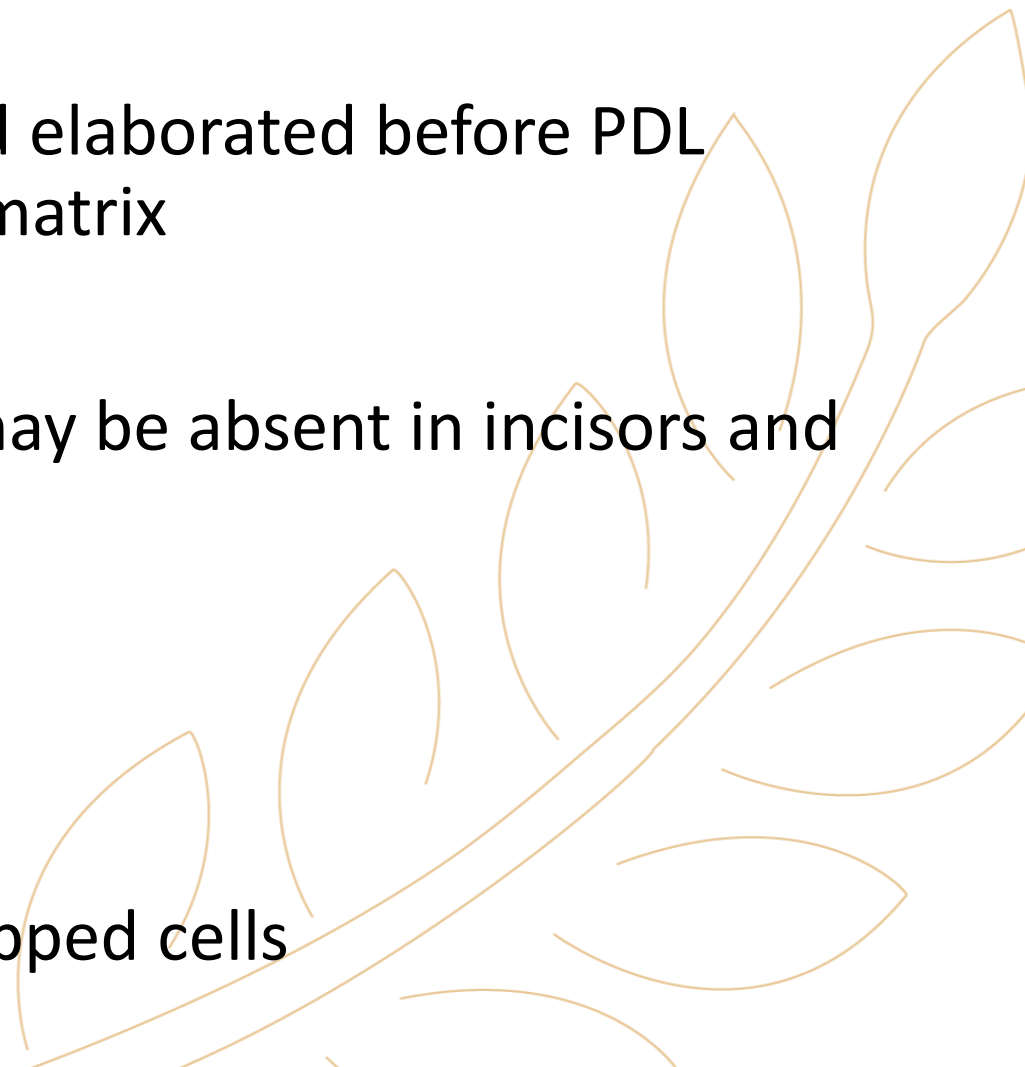


Primary acellular extrinsic fiber cementum

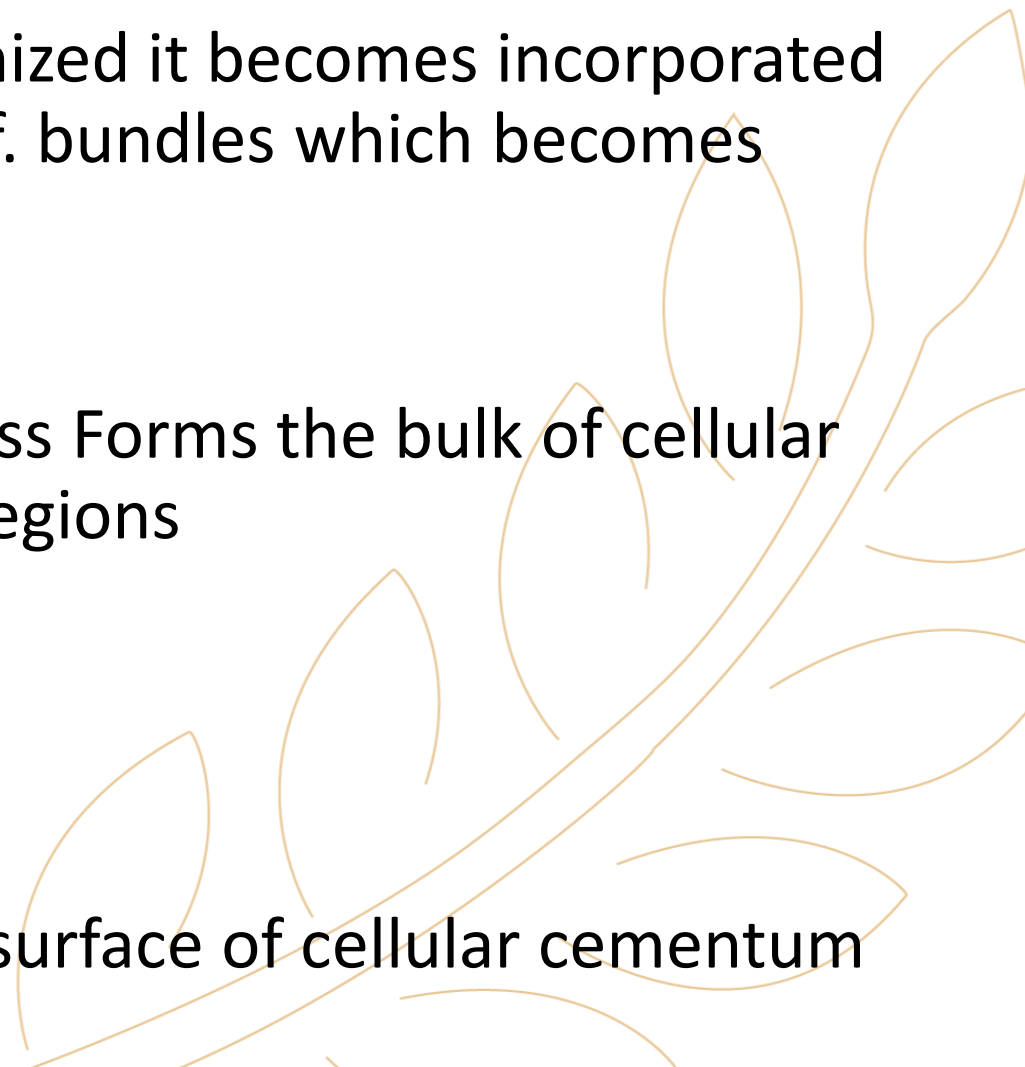
- Forms 90% of acellular cementum
- Matrix: is formed after the PDL fibers connect to the fibrous fring of primary acellular intrinsic fiber cementum.
- Thickness & site: extends form cervical line and covers almost 2/3s of the root forms the bulk of acellular cementum.
- Function: it is the principle tissue of attachment.



Secondary cellular intrinsic fiber cementum

- Matrix: Is a result of cementoblast activity and elaborated before PDL formation the cells are incorporated into the matrix
 - Thickness & site: confined to the apical 1/3 (may be absent in incisors and canines)
 - Function: involved in tooth attachment
 - NB: increase rate of deposition → leads to trapped cells
- 
- A decorative graphic consisting of several stylized, overlapping leaf outlines in a light orange or tan color, positioned in the lower right quadrant of the slide.

Secondary cellular mixed fiber cementum

- Matrix: when PDL fibers are formed and organized it becomes incorporated and cementum is deposited around ligament f. bundles which becomes partially mineralized
 - Thickness and site: Extremely variable thickness Forms the bulk of cellular Cementum, Occupies apical & interradicular regions
 - Function: adaptive tissue
 - NB: cementoid layer is always present on the surface of cellular cementum
- 

Mixed stratified cementum

- Matrix: consists of alternating layers of acellular extrinsic & cellular intrinsic fiber cementum
- Thickness & site: variable, found in apical 1/3 & furcation area
- Function: this alternating of cellular and acellular occurs in response to adaptive need.

